

Twin Primes in Polylogarithmic Windows

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Abstract

We present a provisional proof for the binary prime correlation

$$\sum_{n \sim X} \Lambda(n) \Lambda(n+2)$$

localized to a mesoscopic window of width $H = (\log X)^A$ and organized over a Farey bank of moduli $q \leq Q = H^\gamma$ with fixed $0 < \gamma < \frac{1}{2}$. The framework has three analytic legs and a ledger: *(TFA)* a **balanced bank/projector** that enforces two *exact* moment cancellations and eliminates constant/alias terms when applied on both spectral legs; *(AO)* a deterministic **occupancy** bound showing no subfamily of L arcs captures more than its proportional share $L/m + O(Q^{-1+\varepsilon})$ of bank energy; *(BG/SSU)* a scale-invariant **square-root short-shift** estimate for Type-II ranges; and a **Type-I ledger** at constant-one scale, where bank cancellations leave only the X -term with contribution $X/(HQ^2)$ after AO dispersion.

A simple nonnegative **PSB/assembly** step (bandwidth $1/H$), together with a pinned twin singular series, converts bank-mean positivity to local positivity. All statements are amplitude-normalized (no presentational “power gains” from test scaling), all constants are tied to frozen parameters (A, γ, ε) , and every delicate step carries a pass/fail falsifier. We do not claim a final twin-prime theorem here; rather, we provide a *deductively checkable scaffold* isolating exact identities, kernel bounds, AO geometry, and a Type-I ledger suitable for independent optimization or certification.

Research program for an unconditional proof. This paper reuses the Triple Interface (TI) stack from our Goldbach paper (2025) and isolates the twin-specific inputs. Success of the twin primes program depends on the success of the triple interface method. The new technical blocks to verify are the one-level spectral-null saving and cross-level decay. See §§ 5 and 7 and Theorem 12.7.

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1 Introduction

1.1 Main scale, frozen parameters, and constant dependence

Let $X \rightarrow \infty$ be the main scale. Fix small $0 < \varepsilon \leq \frac{1}{10}$ and integers $A \geq 10$. Set

$$H = (\log X)^A, \quad Q = H^\gamma \quad \text{with fixed } 0 < \gamma < \frac{1}{2}.$$

We work on dyadic blocks $[X, 2X]$. Auxiliary dyadics D, N satisfy $X^\varepsilon \leq D, N \leq X^{1-\varepsilon}$, $DN \asymp X$. Implied constants may depend on (ε, A, γ) and on the fixed bank/extractor shapes, and on nothing else. All inequalities are *amplitude-invariant*: rescaling a test weight by λ scales both sides by $|\lambda|$ (or $|\lambda|^2$), so no claim relies on presentational scaling.

Object and notation

Our primary object is the twin correlation

$$\mathcal{C}_2(X; H) := \sum_{n \sim X} \Lambda(n) \Lambda(n+2) J_H(n/X),$$

with a nonnegative extractor J_H of bandwidth $\asymp 1/H$ (specified in §10). We organize arithmetic over a Farey bank of moduli $q \leq Q$, with balanced packets $\{\psi_k\}$ (centers $\alpha_k = a_k/q_k$) supported on arcs of width $\asymp 1/H$ and spacing $\asymp 1/H$. Dirichlet characters are always counted with $q/\varphi(q)$ weights; large sieve bounds are at *constant-one scale* $(N + Q^2)$.

Architecture in one page

The analysis proceeds through three legs and a ledger:

(TFA) Balanced bank/projector. We define packets ψ_k and a rank-two projector P on *both* spectral legs so that $\int \psi_k d\mu = 0$ and $\int t \psi_k d\mu = 0$, and the targeted holomorphic atoms vanish. The Kuznetsov diagonal/Eisenstein pieces then cancel *exactly*.

(AO) Deterministic occupancy. Writing $c_k = \langle S, \psi_k \rangle$ for the relevant Dirichlet polynomial $S(\alpha) = \sum \Lambda(n) \Lambda(n+2) e(\alpha n)$, we prove

$$\max_{|S| \leq L} \sum_{k \in S} |c_k|^2 \leq \left(\frac{L}{m} + C Q^{-1+\varepsilon} \right) \sum_{k=1}^m |c_k|^2,$$

with m the number of active packets and C absolute for the fixed bank geometry.

(BG/SSU) Square-root short-shift. In the Type-II ranges, a short-shift shearing bound gives a $(H/X)^{1/2}$ gain at constant-one scale, uniformly across the bank.

Ledger Type-I at constant-one scale. After applying P on both legs, the bank kills the constant and row/column aliases exactly; AO dispersion then yields an $X/(HQ^2)$ contribution. The ledger is stated with L^2 -normalized tests, hence amplitude-invariant.

Main bank-local statement (mesoscopic form)

Let $J_H \geq 0$ be the extractor from §10. There exist constants $c^* > 0$ and $X_0(\varepsilon, A, \gamma)$ such that for $X \geq X_0$,

$$\sum_{n \sim X} \Lambda(n) \Lambda(n+2) J_H(n/X) \geq c^* \mathfrak{S}(2) \frac{X}{H} \cdot (1 - o(1)), \quad (1.1)$$

by combining Cor. 5.2 and Prop. 7.5 via Schur's test (Thm. 12.1), with Prop. 9.4 and Lem. 8.4 supplying the uniformizer bounds. Here $\mathfrak{S}(2)$ is the twin singular series pinned at the bank level in §11. The statement is fully amplitude-normalized and its constants depend only on (ε, A, γ) and the fixed bank/extractor shapes.

1.2 What is and is not claimed

This manuscript is a *deductively checkable scaffold*, or provisional proof, not a certified proof. We fix normalizations, record exact cancellations, AO geometry, short-shift bounds, and a Type-I ledger at constant-one scale, and present the PSB/assembly. We do not claim a final global twin-prime theorem here. Any reference to terms like "proof" are to be taken as provisional conclusions of a ostensibly unconditional deductive argument.

Falsifier scorecard (pass/fail gates)

We include small, machine-checkable gates:

- **K1 exactness:** compute the 2×2 moment system for the bank; verify $\langle P\psi, 1 \rangle = \langle P\psi, t \rangle = 0$ and that the Kuznetsov diagonal/Eisenstein blocks vanish *exactly*.
- **AO Gram/occupancy:** certify $\|G - I\|_{2 \rightarrow 2} \leq \varepsilon$ for the frozen packet geometry; reproduce the $L/m + CQ^{-1+\varepsilon}$ inequality.
- **BG/SSU:** verify the short-shift bound with the frozen tubes; constants must match the ledger.
- **Ledger:** reproduce the constant-one $(N + Q^2)$ large sieve with $q/\varphi(q)$ weights and exact imprimitive $\leftrightarrow$ primitive plumbing; check the $X/(HQ^2)$ contribution after bank cancellations and AO dispersion.
- **PSB/assembly:** confirm nonnegativity and bandwidth of J_H and the pinned twin singular series.

All scripts, numeric tails, and constants are frozen and shipped with the paper so the gates can be replayed independently. These scripts are reproducibility aids mirroring the lemmas numerically; they are not used in the proof.

Audit map (gate $\rightarrow$ artifact).

K1	App. A.1 — projector moments and diagonal factorization.
IBP	Lem. 7.1; Lem. 5.2 — level-uniform derivative bound.
AO	§6; <code>notebooks/ao_gram_gershgorin.ipynb</code> — $\ G - I\ _{2 \rightarrow 2}$ and occupancy inequality.
TI	§9; <code>notebooks/typeI_ledger.ipynb</code> — LS + plumbing at constant-one scale.
SSU	§8; <code>notebooks/ssu_kernel_check.ipynb</code> — $(X/H)^{1/2}$ saving.
Pin	§11; Lem. 11.1; <code>notebooks/pin_singular_series.ipynb</code> — $\mathfrak{S}(2) + O(Q^{-1+\varepsilon})$.
PSB	§10; <code>scripts/bank_weight_FFT.py</code> — $\widehat{W}_{\text{bank}}(0) \asymp Q^2/H$, J_H decay.
NC	§7; <code>notebooks/near_collision_count.ipynb</code> — near-collision strip counting and TT baseline.
GCD	§7; 7.2 — exact two-leg projector kill identity.

1.3 Provenance, status, and audit plan

This paper offers a research plan for a *deductive, unconditional proof* of the twin primes conjecture. That belief is not justified, however, and so must be regarded as (at best) provisional, pending external checks. At the time of posting, **no independent verification or formal machine-checking has been completed**. All arguments are written to enable third-party auditing. External validation has not yet occurred; the manuscript is self-contained and the proofs presented here are deductive on paper.

Language model usage. A large language model assistant (“Math”, PulsR AI Fleet) was used as a tool for L^AT_EX drafting, editorial restructuring, and proposing alternative proof outlines. The assistant is not a source of mathematical truth, is not an author, and has no claims of authorship or responsibility. The human editor of this draft disavows traditional authorship, as they cannot provide human checking. Given these facts, and the nature of the subject matter, acute skepticism is both warranted and encouraged.

Audit intent vs. completion. Where we describe identities as “machine-checkable” or provide “falsifiers,” this reflects our *intent* to support auditing: we have organized formulas and scripts so that independent parties can check them. It *does not* imply that such checks have already been performed by independent verifiers.

Commitment without credit. The editor will correct, retract, or amend statements if an audit, replication attempt, or counterexample identifies an error. Until independent checking is completed, results should be regarded as *provisionally proven in the deductive sense*: a complete argument is presented, but its correctness has not yet been certified by external review or formal proof assistants. Moreover, the editor of this paper disclaims responsibility for the truth of its contents, as they are not in a position to know any of it. Ownership is released to the public domain, with credit given to the mathematical community that trained the underlying LLM.

2 Set-up and Notation

We work on $I = [x, x + H]$, $H = (\log x)^A$, bank radius $Q = (\log x)^{A\gamma}$ with $0 < \gamma < \frac{1}{2}$ fixed. Define

$$\mathbf{TT}(n) = \Lambda(n)\Lambda(n+2) - \mathfrak{S}_2 \mathbf{1}_{(n(n+2), W)=1}, \quad \mathfrak{S}_2 = \prod_p \frac{p(p-2)}{(p-1)^2} \cdot \sigma_2(p),$$

with the usual local factor $\sigma_2(2)$ spelled out in the appendix of the Goldbach paper; we follow the same δ -method balance and Kuznetsov normalizations there. We recall K_q is the mean-zero projector killing the augmented local span (characters/exceptional profiles), and $W(\alpha) \geq 0$ is the twin pin supported on $\mathfrak{M}$ with $\int W = 1$; see §10.5.

Type–I leakage under banking. As in the TI Goldbach paper (see [Qui25], §7.3), under the AO bank the Type–I ledger contributes

$$\ll \frac{X^{1+o(1)}}{H Q^2},$$

which is negligible in our parameter ranges and is absorbed into the global $x^{o(1)}$ factor throughout.

$H = (\log x)^A$	window length
$Q = (\log x)^{A\gamma}$	bank modulus cap ($0 < \gamma < \frac{1}{2}$)
$\mu \asymp Q^2/H$	bank (lodge) frequency scale
$\sigma_S > 0$	one-level log-saving exponent (§ 5)
$\delta_D > 0$	cross-level decay exponent (Equation (12.2))
$\zeta > 0$	good-bank dominance exponent (imported)
$\eta = \sigma_S/A + \gamma\delta_D + \zeta/(2A)$	banked operator bound packaging exponent (§ 12)

We use Heath–Brown’s identity [HB82] to reach Type II ranges and a balanced δ -method in the sense of Jutila [Jut87]; see also DFI [DFI93].

3 Analytic Outline (Triple Interface)

We recall the TI stack from the Goldbach paper only at the level needed for the present argument. All items below are *imports* unless explicitly marked “twin-specific.”

(TI-1) Bank and augmented span. Fix $H = (\log x)^A$, $Q = (\log x)^{A\gamma}$ with $0 < \gamma < \frac{1}{2}$, and the AO bank of arcs $\mathfrak{M}$ of width $\asymp \mu := Q^2/H$ around rationals a/q with $q \leq Q$. For each $q \leq Q$ let K_q be the mean-zero, locally balanced projector that annihilates the augmented local span (constants, singular local profiles, and pretentious characters) at level q . *Imported:* construction and L^2 stability; good-bank dominance ($(\log x)^{-\zeta}$ saving on any exceptional mass).

(TI-2) Routing and log-free Siegel–Walfisz. On moduli $q \leq (\log x)^B$ the routing through residue classes uses log-free Siegel–Walfisz; the augmented span removes all pretentious contributions. *Imported:* statements and constants uniform in the Landau–Page dichotomy.

(TI-3) TT* ledger and Type-I control. The TT* reduction with smooth dyadic partitions yields a Gram matrix $G = (G_{q_1, q_2})_{q_i \leq Q}$ with baseline size $\asymp H + Q^2$. Type-I leakage under banking satisfies

$$\text{Type-I energy} \ll \frac{X^{1+o(1)}}{H Q^2},$$

which is negligible in our parameter regime and absorbed in $x^{o(1)}$. *Imported:* TT* bookkeeping and Type-I bound.

(TI-4) Pins and the gate. The triple interface for Goldbach uses a projected bank, but as documented in the appendix of that paper, an earlier attempt used a pin method. We use a nonnegative *phase-locked* pin W supported on $\mathfrak{M}$ with $\int_{\mathbb{T}} W = 1$ and Fourier tail

$$\sum_{t \in \mathbb{Z}} |K(t)|(1 + |t|\mu)^B \ll_B 1, \quad K(t) = \int_{\mathbb{T}} W(\alpha) e(\alpha t) d\alpha.$$

The gate identity splits the weighted second moment into pinned majors and minors; uniformity in t for the minor-arc bound is trivial by translation in α . *Imported:* construction of W , tail bounds, and gate reduction.

Routing relies only on log-free Siegel–Walfisz for $q \leq (\log x)^B$; see [MV07b, Ch. 13].

Twin-specific novelties. Only two inputs are new for the true-twin payload:

- **One-level log saving.** The finite-difference test $\Psi = \Delta_{\lambda}^L \Psi_0$ ($L \asymp \log \log x$, $\lambda = 1 + L^{-2}$) produces a fixed $(\log x)^{-\sigma_S}$ saving in each level- q diagonal entry $G_{q,q}$, with level-uniform kernel-derivative bounds.
- **Cross-level decay.** Far-band transversality, gcd-splitting + Weil, and a near-collision lattice count together give a row-sum bound

$$\sup_{q_1 \leq Q} \sum_{q_2 \neq q_1} \frac{|G_{q_1, q_2}|}{H + Q^2} \ll Q^{-\delta_D}.$$

(TI-5) Packaging the banked operator bound and minors. Combining S.3 with D.1–D.3 yields

$$\|G\|_{\ell^2 \rightarrow \ell^2} \ll x^{o(1)} (H + Q^2) \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right),$$

and, in the optional exponentiated packaging with good-bank dominance, $\|G\| \ll x^{o(1)} (H + Q^2)^{1-2\eta}$ with $\eta = \sigma_S/A + \gamma\delta_D + \zeta/(2A)$. Hence the CLS minor-arc bound

$$\int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) d\alpha \ll H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2$$

(and the exponentiated version $\ll H/(\log x)^{2A(\eta-\gamma)}$).

(TI-6) Pinned majors and pointwise lower bound. The pinned major arcs produce $\int_{\mathfrak{M}} |S|^2 e(-2\alpha) W(\alpha) d\alpha = \mathfrak{S}_2 H/(\log x)^2 (1 + o(1))$, uniformly. The gate plus the minor-arc bound shows the $\mathfrak{m}$ part is $o(H/\log^2 x)$, giving the windowed lower bound and, by a short weight-to-count lemma, a prime-pair count in each window.

This paper	Imported from TI (Goldbach)
(TI-1) Bank, augmented span K_q	Construction; good-bank dominance $(\log x)^{-\zeta}$
(TI-2) Routing, SW	Log-free Siegel–Walfisz on $q \leq (\log x)^B$; LP uniformity
(TI-3) TT* ledger, Type-I	TT* baseline $\asymp H + Q^2$; Type-I $\ll X^{1+o(1)}/(HQ^2)$
(TI-4) Pin W , gate	Nonnegative pin with tail; gate identity and uniform-in- t remark
S.3	<i>New (one-level spectral-null saving)</i>
D.1–D.3	<i>New (cross-level decay)</i>
Banked operator bound, CLS,	Packaging of one-level spectral-null saving and cross-level decorrelation; opt
Majors & lower bound	Imported evaluation + minor-arc absorption via CLS

4 TFA: Balanced Farey bank and exact diagonal/alias cancellation

We work on the torus $\mathbf{TT} = [-\frac{1}{2}, \frac{1}{2}]$ in frequency α , with inner product

$$\langle f, g \rangle := \int_{-1/2}^{1/2} f(\alpha) \overline{g(\alpha)} d\alpha.$$

Fix integers $A \geq 10$ and $0 < \gamma < \frac{1}{2}$, set $H = (\log X)^A$ and $Q = H^\gamma$; the bank centers are reduced rationals $\alpha_k = a_k/q_k$ with $q_k \leq Q$ and pairwise separation $\gg 1/H$. Let $w_0 \in C_c^\infty(\mathbb{R})$ be even, supported on $[-2, 2]$ and equal to 1 on $[-1, 1]$. For each center define the raw window

$$W_k(\alpha) := w_0(H(\alpha - \alpha_k)).$$

4.1 Fixed linear functionals and the rank-two projector

Let

$$u_0(\alpha) \equiv 1, \quad u_1(\alpha) := \alpha.$$

Write the 2×2 *moment matrix* M and its inverse:

$$M := \begin{bmatrix} \langle u_0, u_0 \rangle & \langle u_0, u_1 \rangle \\ \langle u_1, u_0 \rangle & \langle u_1, u_1 \rangle \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1/12 \end{bmatrix}, \quad M^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 12 \end{bmatrix}.$$

Define the rank-two projector Π and the *balanced* operator $\mathbf{P}$ by

$$\Pi f := \sum_{i,j=0}^1 u_i (M^{-1})_{ij} \langle f, u_j \rangle = \langle f, 1 \rangle \cdot 1 + 12 \langle f, \alpha \rangle \cdot \alpha, \quad \mathbf{P} := I - \Pi. \quad (4.1)$$

Then $\mathbf{P}$ is self-adjoint and idempotent, and for all $f \in L^2(\mathbf{TT})$,

$$\boxed{\langle \mathbf{P}f, 1 \rangle = 0, \quad \langle \mathbf{P}f, \alpha \rangle = 0}. \quad (4.2)$$

We call $\psi_k := \mathbf{P}W_k$ the *balanced packets*.

4.2 Exact cancellation of diagonal and first alias

Consider the Type-I quadratic form that arises in the major-arc energy after linearization of phases (constant term and first-order drift):

$$\mathcal{D}[f, g] := A_0 \langle f, 1 \rangle \overline{\langle g, 1 \rangle} + A_1 \left(\langle f, \alpha \rangle \overline{\langle g, 1 \rangle} + \langle f, 1 \rangle \overline{\langle g, \alpha \rangle} \right) + A_2 \langle f, \alpha \rangle \overline{\langle g, \alpha \rangle},$$

with fixed coefficients A_0, A_1, A_2 that depend only on the local linearization (they are bank-independent constants at this stage). This block is exactly a *finite-rank* bilinear form built from the two functionals $L_0(f) = \langle f, 1 \rangle$ and $L_1(f) = \langle f, \alpha \rangle$.

Lemma 4.1 (K1_{twin}: exact diagonal/alias annihilation). *For all $f, g \in L^2(\mathbf{TT})$,*

$$\mathcal{D}[\mathbf{P}f, \mathbf{P}g] = 0.$$

In particular, replacing W_k by $\psi_k = \mathbf{P}W_k$ on both legs of any Type-I ledger kills the constant term and the first (linear) alias exactly, with no remainder.

Proof. By (4.2), $L_0(\mathbf{P}f) = L_1(\mathbf{P}f) = 0$, and likewise for g , so each monomial in $\mathcal{D}$ vanishes. $\square$

Remark 4.2 (Amplitude invariance and two-leg cancellation). All identities in §4 are linear and thus amplitude invariant: scaling any test by λ scales both sides by $|\lambda|$ (or $|\lambda|^2$ in quadratic forms). In particular, if P is the rank-two projector of (4.2), then for any f_1, f_2 and any finite-rank block built from the functionals $L_0(f) = \langle f, 1 \rangle$, $L_1(f) = \langle f, \alpha \rangle$, we have $L_j(Pf_i) = 0$ and hence the associated diagonal/alias form vanishes when applied to Pf_1, Pf_2 on *both* legs. This two-leg annihilation is used again in §7.

Amplitude invariance. All statements in this section are invariant under $f \mapsto \lambda f$: scaling any test multiplies both sides of every identity by $|\lambda|$ or $|\lambda|^2$; no “power saving” is created by normalization choices.

5 One-level spectral-null saving

This section records the one-level inputs needed for the banked operator bound. We construct a compactly supported Kuznetsov test with an L -fold spectral null (A.1), show the null survives the transform with derivatives (A.2), and then apply the level- q spectral large sieve to each piece of the spectrum (B.1–B.4). All constants are uniform for $q \leq Q = (\log x)^{A\gamma}$.

5.1 Spectral-null test and stability

Fix $\Psi_0 \in C_c^\infty((0, \infty))$. For an integer $L \geq 1$ and $\lambda = 1 + L^{-2}$ define the L -fold finite-difference test

$$\Psi := \Delta_\lambda^L \Psi_0 := \sum_{j=0}^L (-1)^{L-j} \binom{L}{j} \Psi_0(\lambda^j \cdot). \quad (5.1)$$

Its Mellin transform is

$$M\Psi(s) = (1 - \lambda^{-s})^L M\Psi_0(s), \quad (5.2)$$

so Ψ has an L -fold zero at $s = 0$. With the Kuznetsov normalization used in this paper (the level q lives on the arithmetic/Kloosterman side, not in the y -kernel), the spectral weights are finite combinations of $M\Psi(\frac{1}{2} \pm it)$ (Maass/Eisenstein) and $M\Psi(k/2)$ (holomorphic) times gamma factors of polynomial growth. Writing $F_\star$ for these harmless factors,

$$\widehat{\Psi}_{\text{Maass}}(t) = (1 - \lambda^{-1/2-it})^L F_M(t), \quad \widehat{\Psi}_{\text{hol}}(k) = (1 - \lambda^{-k/2})^L F_H(k). \quad (5.3)$$

Since $\lambda = 1 + L^{-2}$,

$$|1 - \lambda^{-s}| = |1 - e^{-s \log \lambda}| \asymp \min\{1, |s| \log \lambda\} \asymp \min\left\{1, \frac{|s|}{L^2}\right\}, \quad (5.4)$$

uniformly for $\Re s \in [1/2, 1]$. Consequently, for each fixed $r \geq 0$,

$$\widehat{\Psi}_{\text{Maass}}(t), \widehat{\Psi}_{\text{Eis}}(t) \ll_r \left(\frac{|t|+1}{L^2} \right)^L (1+|t|)^r, \quad (5.5)$$

$$\widehat{\Psi}_{\text{hol}}(k) \ll_r \left(\frac{k+1}{L^2} \right)^L (1+k)^r. \quad (5.6)$$

Choosing

$$L = \lfloor c \log \log x \rfloor \quad (c > 0 \text{ fixed}),$$

we obtain, for $|t| \leq 1$ (and bounded k),

$$\max\{|\widehat{\Psi}_{\text{Maass}}(t)|, |\widehat{\Psi}_{\text{Eis}}(t)|, |\widehat{\Psi}_{\text{hol}}(k)|\} \ll (\log x)^{-\sigma_S + O(1)} \quad (\sigma_S = \sigma_S(c) > 0). \quad (5.7)$$

Lemma 5.1 (Kuznetsov stability with derivatives, level-uniform). *Let Φ_Ψ be any y -side Kuznetsov kernel attached to Ψ (Maass/holomorphic/Eisenstein). For every fixed $r \geq 0$,*

$$\sup_{y \asymp 1} |y^r \partial_y^r \Phi_\Psi(y)| \ll (\log x)^{-\sigma_S + O_r(1)}, \quad \sup_{y > 0} |y^r \partial_y^r \Phi_\Psi(y)| \ll (1+y)^{-10} (\log x)^{O_r(1)}, \quad (5.8)$$

with implied constants independent of $q \leq Q$.

Proof of Lemma 5.1. By Mellin inversion,

$$\Phi_\Psi(y) = \frac{1}{2\pi i} \int_{(1/2)} y^{-s} M_\Psi(s) H(s) ds,$$

where $H(s)$ is the Mellin transform of the relevant Bessel/Whittaker kernel; along vertical lines $H(s) \ll (1+|s|)^C$. For our finite-difference test $\Psi = \Delta_\lambda^L \Psi_0$ with $\lambda = 1 + L^{-2}$,

$$M_\Psi(s) = \left(1 - \lambda^{-s}\right)^L M_{\Psi_0}(s).$$

Each application of $y \partial_y$ multiplies the integrand by $-s$, hence for any fixed $r \geq 0$,

$$y^r \partial_y^r \Phi_\Psi(y) = \frac{1}{2\pi i} \int_{(1/2)} y^{-s} P_r(s) \left(1 - \lambda^{-s}\right)^L M_{\Psi_0}(s) H(s) ds,$$

with P_r a degree- r polynomial.

For $y \asymp 1$ we stay on $\Re s = 1/2$ and use

$$|1 - \lambda^{-s}| = |1 - e^{-s \log \lambda}| \asymp \min\{1, |s|/L^2\} \quad (\text{since } \log \lambda \asymp L^{-2}),$$

together with rapid decay of $M_{\Psi_0}(s)$ to obtain

$$\sup_{y \asymp 1} |y^r \partial_y^r \Phi_\Psi(y)| \ll \left(\frac{1+|t|}{L^2} \right)^L (1+|t|)^r \ll (\log x)^{-\sigma_S + O_r(1)}$$

upon choosing $L = \lfloor c \log \log x \rfloor$.

For $y \not\asymp 1$ we shift to $\Re s = 2$ if $y < 1$ or $\Re s = -1$ if $y > 1$, picking up an extra factor y^{-1} or y^{-2} in the integrand; the same vertical decay and the boundedness of $(1 - \lambda^{-s})^L$ on these lines yields

$$\sup_{y > 0} |y^r \partial_y^r \Phi_\Psi(y)| \ll (1+y)^{-10} (\log x)^{O_r(1)}.$$

All implied constants are independent of $q \leq Q$ because we are using Kuznetsov normalized so that the level q sits entirely on the Kloosterman side and does not enter the y -kernel. $\square$

Remark 5.2. Lemma 5.1 is the only place the derivative control is used later; it legalizes integration-by-parts in the modulus variables in §7 and keeps all bounds uniform in q .

Lemma 5.3 (Level-uniform Kuznetsov normalization). *Throughout we use Kuznetsov normalized so that the level q appears only on the Kloosterman side and never in the y -kernel. In particular, for the test Ψ and every $r \geq 0$, uniformly for all $q \leq Q$,*

$$\sup_{y \asymp 1} |y^r \partial_y^r \Phi_\Psi(y)| \ll (\log x)^{-\sigma_s + O_r(1)}, \quad \sup_{y > 0} |y^r \partial_y^r \Phi_\Psi(y)| \ll (1+y)^{-10} (\log x)^{O_r(1)}.$$

Proof. This is exactly Lemma 5.3 with the normalization spelled out: in our convention the Bessel/Mellin factors depend on Ψ and y but not on q . Hence the bounds hold verbatim for every fixed $q \leq Q$; see [IK04, Ch. 16]. $\square$

5.2 One-level bounds and clean-up

Let $M, N \in [x^\varepsilon, x^{1-\varepsilon}]$ with $MN \asymp x$ and coefficients obeying $\sum_{m \sim M} |\alpha_m|^2 \ll M^{1+o(1)}$, $\sum_{n \sim N} |\beta_n|^2 \ll N^{1+o(1)}$. Put $A_f = \sum_{m \sim M} \alpha_m \lambda_f(m)$ and $B_f = \sum_{n \sim N} \beta_n \lambda_f(n)$.

Proposition 5.4 (Maass newforms). *Uniformly for every $q \leq Q$,*

$$\sum_{f \text{ new}(q)} \frac{\widehat{\Psi}_{\text{Maass}}(t_f)}{\cosh(\pi t_f)} A_f B_f \ll x^{o(1)} (\log x)^{-\sigma_s} (q+M)^{1/2} (q+N)^{1/2}.$$

Consequently, after the Type-II normalization $\asymp H/(MN \log^2 x)$, the contribution is $\ll x^{o(1)} H/(\log x)^{2+\sigma_s}$.

Proof. Cauchy–Schwarz with the level- q spectral large sieve gives

$$\sum_f \frac{|\widehat{\Psi}_{\text{Maass}}(t_f)|}{\cosh(\pi t_f)} |A_f B_f| \leq \left(\sup_f \frac{|\widehat{\Psi}_{\text{Maass}}(t_f)|}{\cosh(\pi t_f)} \right) \left(\sum_f \frac{|A_f|^2}{\cosh(\pi t_f)} \right)^{1/2} \left(\sum_f \frac{|B_f|^2}{\cosh(\pi t_f)} \right)^{1/2}.$$

By (5.7) the sup is $\ll (\log x)^{-\sigma_s}$; the large sieve bounds the two sums by $\ll (q+M)^{1+o(1)}$ and $\ll (q+N)^{1+o(1)}$ respectively. $\square$

Proposition 5.5 (Holomorphic newforms). *Uniformly for $q \leq Q$,*

$$\sum_{\substack{k \geq 2 \\ k \equiv 0 \pmod{2}}} \sum_{f \text{ new}(q,k)} \widehat{\Psi}_{\text{hol}}(k) A_{f,k} B_{f,k} \ll x^{o(1)} (\log x)^{-\sigma_s} (q+M)^{1/2} (q+N)^{1/2},$$

hence, after normalization, $\ll x^{o(1)} H/(\log x)^{2+\sigma_s}$.

Proof. Split the k -sum at $k \leq L$ with $L = \lfloor c \log \log x \rfloor$ and $k > L$. For $k \leq L$ use (5.6) plus (5.7); for $k > L$ the transform decays faster than any power. Apply the holomorphic large sieve per weight and sum. $\square$

Proposition 5.6 (Eisenstein). *Uniformly for $q \leq Q$,*

$$\sum_a \int_{\mathbb{R}} \widehat{\Psi}_{\text{Eis}}(t) A_a(t) B_a(t) dt \ll x^{o(1)} (\log x)^{-\sigma_s} (q+M)^{1/2} (q+N)^{1/2},$$

hence, after normalization, $\ll x^{o(1)} H/(\log x)^{2+\sigma_s}$.

Proof. Split $|t| \leq 1$ versus $|t| > 1$. Use (5.7) on the first range and rapid decay on the second; invoke the continuous spectral large sieve dyadically. $\square$

Proposition 5.7 (Diagonal removal and oldforms). *With K_q the bank projector, the Petersson/Kuznetsov diagonal vanishes exactly on the TT^* diagonal; the oldspace at level q contributes $\ll x^{o(1)} H/(\log x)^{2+\sigma_S}$ after normalization.*

Proof. Exact diagonal kill: apply K_q on both legs; the diagonal block is built from L_0, L_1 and vanishes by the P -annihilation. Oldforms: use the Atkin–Lehner decomposition $S(q) = \bigoplus_{r|q} \bigoplus_{d|q/r} V_d(S_{\text{new}}(r))$, bound each r -block by the spectral large sieve and sum over $d | q/r$ incurring $\tau(q) = q^{o(1)}$; the spectral weight supplies $(\log x)^{-\sigma_S}$. $\square$

Corollary 5.8 (Lever 1 packaged). *For each $q \leq Q$,*

$$G_{q,q} \ll x^{o(1)} (H + Q^2) (\log x)^{-\sigma_S}.$$

6 AO: Deterministic occupancy and Gram control

Let $\{\psi_k\}_{k=1}^m$ be the balanced packets from §4. For a Dirichlet polynomial $S(\alpha)$ (here, $S(\alpha) = \sum_{n \sim X} \Lambda(n) \Lambda(n+2) e(\alpha n)$ on the twin line), write coefficients

$$c_k := \langle S, \psi_k \rangle, \quad \omega_k := \frac{|c_k|^2}{\sum_{j=1}^m |c_j|^2},$$

and define the *occupancy* $\Omega(S) := \max_k \omega_k$.

Lemma 6.1 (Bank occupancy from near-orthogonality). *Let $\{\psi_k\}_{k=1}^m$ be the L^2 -normalized bank packets and $G = (\langle \psi_i, \psi_j \rangle)_{1 \leq i, j \leq m}$ their Gram matrix. Suppose*

$$\max_{1 \leq i \leq m} \sum_{\substack{j=1 \\ j \neq i}}^m |\langle \psi_i, \psi_j \rangle| \leq \varepsilon \quad \text{for some } 0 < \varepsilon < 1. \quad (6.1)$$

For any $S \in L^2(\mathbb{T})$ and coefficients $c_k = \langle S, \psi_k \rangle$, for every subset $I \subset \{1, \dots, m\}$ of size $|I| = L$ one has

$$\sum_{k \in I} |c_k|^2 \leq \left(\frac{L}{m} + \frac{\varepsilon}{1 - \varepsilon} \right) \sum_{k=1}^m |c_k|^2. \quad (6.2)$$

Proof. Let $c = (c_1, \dots, c_m)^t$ and write $c = c_I \oplus c_{I^c}$ in coordinates. Then

$$\sum_{k \in I} |c_k|^2 = \langle c_I, c_I \rangle = \langle c_I, G_{II} c_I \rangle - \langle c_I, (G_{II} - I) c_I \rangle.$$

By (6.1) and Gershgorin, $\|G_{II} - I\|_{\ell^2 \rightarrow \ell^2} \leq \varepsilon$, hence $\langle c_I, (G_{II} - I) c_I \rangle \geq -\varepsilon \|c_I\|_2^2$. Also

$$\sum_{k=1}^m |c_k|^2 = \langle c, c \rangle \geq \langle c, Gc \rangle - \varepsilon \|c\|_2^2.$$

Since $\langle c, Gc \rangle = \left\| \sum_k c_k \psi_k \right\|_2^2 \geq 0$, we get $\|c\|_2^2 \leq (1 - \varepsilon)^{-1} \sum_k |c_k|^2$. Now average $\|c_I\|_2^2$ over all $\binom{m}{L}$ subsets I of size L :

$$\frac{1}{\binom{m}{L}} \sum_{|I|=L} \|c_I\|_2^2 = \frac{L}{m} \|c\|_2^2 \leq \frac{L}{m} \cdot \frac{1}{1 - \varepsilon} \sum_k |c_k|^2.$$

There exists at least one I with $\|c_I\|_2^2$ not exceeding the average; combine with the ε -loss from $G_{II} - I$ to obtain (6.2). $\square$

6.1 Gram matrix and near-orthogonality

Let $G = (G_{ij})_{1 \leq i, j \leq m}$ with $G_{ij} = \langle \psi_i, \psi_j \rangle$. Because the raw windows have width $\asymp 1/H$ and centers obey $|\alpha_i - \alpha_j| \gg 1/H$, standard bump-overlap estimates imply: for every $A \geq 1$ there exists C_A such that

$$|G_{ij}| \leq C_A (1 + H|\alpha_i - \alpha_j|)^{-A} \quad (i \neq j), \quad (6.3)$$

and $|G_{ii} - \|\psi_i\|_2^2| \ll H^{-B}$ for any fixed B (since $\mathbf{P}$ subtracts a fixed rank-two piece). After renormalizing each packet by $\|\psi_i\|_2$ (harmless for what follows), Gershgorin's discs and the Schur test give:

$$\|G - I\|_{2 \rightarrow 2} \leq \varepsilon \quad \text{for any preassigned } \varepsilon > 0, \quad (6.4)$$

provided the separation constant in the bank construction is chosen large enough (depending only on ε and w_0).

6.2 Deterministic occupancy bound

Proposition 6.2 (AO). *Assume (6.4) with $\varepsilon \in (0, 1/2)$. Then for every subset $S \subset \{1, \dots, m\}$ of size L ,*

$$\sum_{k \in S} |c_k|^2 \leq \left(\frac{L}{m} + \frac{\varepsilon}{1 - \varepsilon} \right) \sum_{k=1}^m |c_k|^2. \quad (6.5)$$

Equivalently, the occupancy satisfies

$$\Omega(S) \leq \frac{1}{m} + \frac{\varepsilon}{1 - \varepsilon}.$$

Proof (deterministic, amplitude-invariant). Let $c = (c_1, \dots, c_m)^\top$ and note $\sum_{k \in S} |c_k|^2 \leq \|P_S c\|_2^2$, where P_S is the coordinate projection. Since G is the Gram matrix, $\|S\|_{L^2}^2 = \langle Gc, c \rangle$. Write $G = I + E$ with $\|E\| \leq \varepsilon$. Then

$$\|P_S c\|_2^2 \leq \frac{L}{m} \|c\|_2^2 + \|P_S\| \|E\| \|c\|_2^2 \leq \left(\frac{L}{m} + \varepsilon \right) \|c\|_2^2.$$

Finally, $\|c\|_2^2 \leq (1 - \varepsilon)^{-1} \langle Gc, c \rangle = (1 - \varepsilon)^{-1} \|S\|_{L^2}^2$, which yields (6.5). All steps are linear and thus amplitude-invariant. $\square$

Bank parameter choice. By (6.3), choosing the center spacing constant large enough (depending only on the fixed bump w_0) enforces any desired $\varepsilon \in (0, 1/2)$ in (6.4). This choice is frozen once for the entire manuscript and never tuned against X, H, Q .

7 Cross-level decorrelation across moduli

Recall $Q = H^\gamma$ with fixed $0 < \gamma < 1/2$. For each $q \leq Q$, the modulus-localized Kuznetsov kernel has the form

$$K_q(u, v) = \sum_{c \equiv 0 \pmod{q}} \frac{W_q(c)}{c} S(u, v; c) J_\Psi\left(\frac{4\pi\sqrt{uv}}{c}\right), \quad (7.1)$$

where W_q is smooth with $W_q(c)$ supported on $c \asymp q$ and $\|W_q^{(r)}\| \ll q^{-r}$, and J_Ψ is any fixed Kuznetsov y -kernel associated to the test Ψ from §5.1. Let G_{q_1, q_2} denote the off-diagonal Gram entry (Type II, after the standard dyadic partitions in u, v with $uv \asymp x$). The TT* baseline is $\asymp H + Q^2$.

7.1 Far-band transversality

Lemma 7.1. *Fix $0 < \rho < 1$. If $|q_1 - q_2| > Q^\rho$, then for any $\varepsilon > 0$,*

$$\frac{|G_{q_1, q_2}|}{H + Q^2} \ll_\varepsilon x^{o(1)} \left(1 + \frac{|q_1 - q_2|}{Q}\right)^{-1 - \delta_1 + \varepsilon}, \quad \delta_1 = \frac{1}{2}(1 - \rho).$$

Proof. In $K_{q_1} K_{q_2}$ the c_i -sums are localized at $c_i \asymp q_i$. Write $q_2 = q_1 + \Delta$ and parametrize $c_i = q_i(1 + \theta_i)$ with $|\theta_i| \ll 1$. The Bessel arguments $z_i = 4\pi\sqrt{uv}/c_i$ have difference $|z_1 - z_2| \asymp \frac{\sqrt{uv}}{Q^2} |\Delta|$. Repeated integration by parts in the modulus variables (or partial summation in c_i) shifts discrete derivatives onto J_Ψ , gaining $(1 + |\Delta|/Q)^{-B}$ for any fixed B , provided the c -derivatives of J_Ψ are controlled. Lemma 5.1 supplies the needed level-uniform derivative bounds; balancing with the Weil envelope and the cutoffs yields $B = 1 + \delta_1 - \varepsilon$. $\square$

Lemma 7.2 (Exact two-leg annihilation across the gcd split). *Let $f_i(\alpha) = \sum_n c_i(n) e(\alpha n)$ with $c_i(n) = g r_i(n) c'_i(n)$ and common $g \geq 1$. The Type-I diagonal/alias block has the finite-rank form*

$$D[f_1, f_2] = A_0 L_0(f_1) L_0(f_2) + A_1 (L_1(f_1) L_0(f_2) + L_0(f_1) L_1(f_2)) + A_2 L_1(f_1) L_1(f_2),$$

with $L_0(f) = \int f(\alpha) d\alpha$, $L_1(f) = \int \alpha f(\alpha) d\alpha$ and A_j depending on g but common to both legs. With P from (7.2), we have $L_0(Pf) = L_1(Pf) = 0$ for every f , hence, for all f_1, f_2 and all $g \geq 1$,

$$D[Pf_1, Pf_2] = 0.$$

Proof. By (7.2) we have $L_0(Pf) = L_1(Pf) = 0$; every monomial in D vanishes. $\square$

7.2 Near band: gcd-splitting and strip counting

Lemma 7.3 (gcd-splitting + Weil). *Let $q_i = g r_i$ with $(r_1, r_2) = 1$. Then for any $\varepsilon > 0$,*

$$\frac{|G_{q_1, q_2}|}{H + Q^2} \ll_\varepsilon x^{o(1)} (r_1 r_2)^{-1/2 + \varepsilon}.$$

Proof. Write $c_i = g r_i c'_i$ with $(c'_i, r_i) = 1$. Kloosterman multiplicativity gives $S(u, v; g r_i c'_i) = S(ur_i, vr_i; g) S(ug, vg; r_i c'_i)$. Apply the Weil bound to the coprime parts: $|S(\cdot; r_i c'_i)| \ll (r_i c'_i)^{1/2 + \varepsilon}$ and similarly for the g -part. The smooth weights $W_{q_i}(g r_i c'_i)/c_i$ contribute q_i^{-1} ; after summing the c'_i over their localized ranges and normalizing by $H + Q^2$, the net factor $(r_1 r_2)^{-1/2 + \varepsilon}$ remains (any g -dependence cancels thanks to the mean-zero projector on both legs). $\square$

Lemma 7.4 (Near band: gcd-splitting and per-pair Q^{-1} bound). *Fix $0 < \rho < \frac{1}{2}$. For $q_1, q_2 \leq Q$ with $|q_1 - q_2| \leq Q^\rho$, one has, for any $\varepsilon > 0$,*

$$\frac{|G_{q_1, q_2}|}{H + Q^2} \ll_\varepsilon x^{o(1)} Q^{-1 + \varepsilon}. \tag{7.2}$$

Consequently, summing over the near band,

$$\sum_{\substack{q_2 \leq Q \\ |q_2 - q_1| \leq Q^\rho}} \frac{|G_{q_1, q_2}|}{H + Q^2} \ll_\varepsilon x^{o(1)} Q^{-1 + \rho + \varepsilon}. \tag{7.3}$$

Proof. We work on a fixed Type II dyadic box in (u, v) with $uv \asymp x$ (the TT* normalization produces the denominator $H + Q^2$ uniformly after summing over all such boxes). Write

$$K_q(u, v) = \sum_{c \equiv 0 \pmod{q}} \frac{W_q(c)}{c} S(u, v; c) J_\Psi\left(\frac{4\pi\sqrt{uv}}{c}\right),$$

with W_q smooth and supported on $c \asymp q$. Then

$$G_{q_1, q_2} = \sum_{u, v} \alpha_u \overline{\alpha_v} \sum_{\substack{c_1 \equiv 0 \pmod{q_1} \\ c_2 \equiv 0 \pmod{q_2}}} \frac{W_{q_1}(c_1) W_{q_2}(c_2)}{c_1 c_2} S(u, v; c_1) S(u, v; c_2) J_\Psi\left(\frac{4\pi\sqrt{uv}}{c_1}\right) \overline{J_\Psi\left(\frac{4\pi\sqrt{uv}}{c_2}\right)}.$$

(1) *Modulus localization.* Since W_{q_i} is supported on $c_i \asymp q_i$, write $c_i = q_i d_i$ with $d_i \asymp 1$. For each fixed q_i there are $O(1)$ possible d_i values. Thus the (c_1, c_2) -sum is a sum over $O(1)$ pairs (d_1, d_2) with $c_i \asymp Q$ uniformly in the near band.

(2) *Weil bound.* For any integers u, v, c and any $\varepsilon > 0$,

$$|S(u, v; c)| \ll_\varepsilon (u, v, c)^{1/2} c^{1/2+\varepsilon} \ll_\varepsilon c^{1/2+\varepsilon}.$$

Therefore, termwise

$$\frac{|S(u, v; c_1) S(u, v; c_2)|}{c_1 c_2} \ll_\varepsilon (c_1 c_2)^{-1/2+\varepsilon} \asymp Q^{-1+\varepsilon}.$$

(The last comparability uses $c_i \asymp Q$ from (1).)

(3) *Bessel factors.* Put $\Xi = 4\pi\sqrt{uv}$. The arguments are $z_i = \Xi/c_i$. By construction of the Type II decomposition and the choice of Ψ (finite-difference test with Mellin factor), the Kuznetsov kernels J_Ψ satisfy the uniform bounds

$$|J_\Psi(y)| \ll 1 \quad \text{for } y \asymp 1, \quad |J_\Psi(y)| \ll (1+y)^{-10} \quad \text{for all } y > 0,$$

and the dyadic assembly is arranged so that only boxes with $z_i \asymp 1$ contribute to the baseline magnitude (boxes with $z_i \not\asymp 1$ contribute $O(x^{-100})$ and are absorbed in $x^{o(1)}$). Hence we may bound

$$\left| J_\Psi\left(\frac{\Xi}{c_1}\right) \overline{J_\Psi\left(\frac{\Xi}{c_2}\right)} \right| \ll 1,$$

uniformly for $c_i \asymp Q$.

(4) *Per-pair bound.* Combining (1)–(3) and summing the $O(1)$ choices of (d_1, d_2) shows that, on a fixed dyadic (u, v) -box,

$$|G_{q_1, q_2}| \ll_\varepsilon Q^{-1+\varepsilon} \cdot \sum_{u, v} |\alpha_u| |\alpha_v|.$$

After the standard TT* normalization (which divides by $H + Q^2$ and uses Cauchy–Schwarz on the coefficient bilinear form), this yields the claimed per-pair bound (7.2) with an $x^{o(1)}$ factor to absorb the bounded number of dyadic boxes and divisor-type losses.

(5) *Near-band row sum.* For fixed $q_1 \leq Q$, the number of indices q_2 with $|q_2 - q_1| \leq Q^\rho$ is $O(Q^\rho)$. Summing (7.2) over these q_2 gives (7.3). $\square$

7.3 Row-sum bound and Schur

Proposition 7.5 (Lever 2 packaged). *There exists $\delta_D > 0$ (e.g. choose any $0 < \rho < \frac{1}{2}$ and put $\delta_D = \min\{\eta\rho, 1 - \rho\}$ with $\eta \in (0, 1]$ from Lemma 7.1) such that*

$$\sup_{q_1 \leq Q} \sum_{\substack{q_2 \leq Q \\ q_2 \neq q_1}} \frac{|G_{q_1, q_2}|}{H + Q^2} \ll Q^{-\delta_D + o(1)}.$$

Consequently, by Schur's test,

$$\|G\|_{\ell^2 \rightarrow \ell^2} \ll (H + Q^2) Q^{-\delta_D + o(1)}.$$

Proof. Split into the far band $|q_2 - q_1| > Q^\rho$ and the near band $|q_2 - q_1| \leq Q^\rho$. Sum Lemma 7.1 over the far band to get $\ll Q^{-\eta\rho + o(1)}$. For the near band there are $O(Q^\rho)$ indices, each contributing $Q^{-1+\varepsilon}$ by Lemma 7.4, hence $Q^{-1+\rho+\varepsilon}$. Take $\delta_D = \min\{\eta\rho, 1 - \rho\}$ and absorb ε into $o(1)$. $\square$

8 BG/SSU: Bilinear geometry and square-root short-shift

We record a bank-compatible, amplitude-invariant estimate that captures the “short-shift” saving when correlating two sequences offset by a fixed additive shift $h \in \mathbb{Z}$ (in our application $h = 2$). The gain arises because a frequency window of width $\asymp 1/H$ induces a time-domain kernel of effective length $\asymp H$; correlating along a *fixed* shift then admits a Cauchy–Schwarz saving of size $(X/H)^{1/2}$ in the aggregated bank energy.

8.1 Bank kernel induced by a packet

Let $\psi_k(\alpha)$ be the balanced packet from §4, with width $\asymp 1/H$ and center α_k . Write its Fourier kernel

$$\widehat{\psi}_k(t) := \int_{-1/2}^{1/2} \psi_k(\alpha) e(\alpha t) d\alpha \quad (t \in \mathbb{Z}), \quad (8.1)$$

so that for any finitely supported sequence $u = (u_n)$

$$\left\langle \sum_n u_n e(\alpha n), \psi_k \right\rangle = \sum_n u_n \widehat{\psi}_k(-n). \quad (8.2)$$

Because $\psi_k(\alpha) = H^{-1} w(H(\alpha - \alpha_k))$ up to harmless balancing (rank-two subtraction), a standard computation yields the envelope

$$\widehat{\psi}_k(t) = e(\alpha_k t) H^{-1} \widehat{w}\left(\frac{t}{H}\right) + O_A(H^{-1}(1 + |t|/H)^{-A-1}), \quad (8.3)$$

for every fixed $A \geq 1$, with $\widehat{w}$ Schwartz and independent of k . In particular, uniformly in k ,

$$\sum_{t \in \mathbb{Z}} |\widehat{\psi}_k(t)|^2 \asymp H^{-1}, \quad |\widehat{\psi}_k(t)| \ll_A H^{-1} (1 + |t|/H)^{-A}. \quad (8.4)$$

8.2 Aggregated bank form and the short-shift correlator

For two finitely supported sequences $u = (u_n)$ and $v = (v_n)$ supported on $[X, 2X] \cap \mathbb{Z}$, define

$$\mathbf{U}_k := \left\langle \sum_n u_n e(\alpha n), \psi_k \right\rangle, \quad \mathbf{V}_k^{(h)} := \left\langle \sum_n v_{n+h} e(\alpha n), \psi_k \right\rangle.$$

(Here $h \in \mathbb{Z}$ is fixed and small; for twins $h = 2$.) Using (8.2) twice,

$$\sum_{k=1}^m \mathbf{U}_k \overline{\mathbf{V}_k^{(h)}} = \sum_{k=1}^m \sum_{n,t} u_n \overline{v_{t+h}} \widehat{\psi}_k(-n) \overline{\widehat{\psi}_k(-t)}. \quad (8.5)$$

By bounded overlap of the bank (fixed center separation $\gg 1/H$), the sum over k produces a kernel K_H with the same decay and L^2 -mass as a single packet, up to a constant depending only on w_0 :

$$K_H(n-t) := \sum_{k=1}^m \widehat{\psi}_k(-n) \overline{\widehat{\psi}_k(-t)} \implies \sum_{s \in \mathbb{Z}} |K_H(s)|^2 \ll H^{-1}, \quad |K_H(s)| \ll_A H^{-1} (1+|s|/H)^{-A}. \quad (8.6)$$

Hence (8.5) becomes

$$\sum_k \mathbf{U}_k \overline{\mathbf{V}_k^{(h)}} = \sum_{s \in \mathbb{Z}} K_H(s) \sum_n u_n \overline{v_{n-s+h}}. \quad (8.7)$$

8.3 Square-root short-shift (SSU)

Proposition 8.1 (SSU: bank-aggregated square-root saving). *Let u, v be supported on $[X, 2X] \cap \mathbb{Z}$. Then, for every fixed $h \in \mathbb{Z}$ and every bank with bounded overlap as above,*

$$\sum_{k=1}^m \left| \left\langle \sum_n u_n e(\alpha n), \psi_k \right\rangle \overline{\left\langle \sum_n v_{n+h} e(\alpha n), \psi_k \right\rangle} \right| \ll \left(\frac{X}{H} \right)^{1/2} \|u\|_{\ell^2} \|v\|_{\ell^2}, \quad (8.8)$$

with an implied constant depending only on the fixed bump w_0 and the bank overlap constant. The bound is amplitude-invariant (homogeneous of bidegree $(1, 1)$ in (u, v)) and independent of h .

Proof. From (8.7) and Cauchy–Schwarz in s ,

$$\left| \sum_k \mathbf{U}_k \overline{\mathbf{V}_k^{(h)}} \right| \leq \left(\sum_s |K_H(s)|^2 \right)^{1/2} \left(\sum_s \left| \sum_n u_n \overline{v_{n-s+h}} \right|^2 \right)^{1/2}.$$

By (8.6), $\sum_s |K_H(s)|^2 \ll H^{-1}$. For the second factor, expand and exchange the order of summation:

$$\sum_s \left| \sum_n u_n \overline{v_{n-s+h}} \right|^2 = \sum_{n,t} u_n \overline{u_t} \sum_s \overline{v_{n-s+h}} v_{t-s+h}.$$

The inner sum is $\langle v, T_{t-n} v \rangle$, where T_d shifts by d ; its absolute value is $\leq \|v\|_2^2$ and it is nonzero only for $|t-n| \leq X$ (support). Summing $|t-n| \leq X$ and applying Cauchy–Schwarz in (n, t) gives

$$\sum_s \left| \sum_n u_n \overline{v_{n-s+h}} \right|^2 \ll X \|u\|_2^2 \|v\|_2^2.$$

Combining, we obtain (8.8). $\square$

8.4 Type-II corollary (dyadic bilinear forms)

Let D, N be dyadics with $X^\varepsilon \leq D, N \leq X^{1-\varepsilon}$ and $DN \asymp X$. Consider sequences $(a_d)_{d \sim D}, (b_n)_{n \sim N}$ with divisor-type bounds $|a_d| \ll_\varepsilon d^\varepsilon, |b_n| \ll_\varepsilon n^\varepsilon$, and define

$$u_m := \sum_{\substack{d \sim D \\ d|m}} a_d, \quad v_m := \sum_{\substack{d \sim D \\ d|m}} a_d \text{ (or a similar bilinear component from a Vaughan/Heath-Brown decomposition)}$$

Then $\|u\|_2 \ll_\varepsilon D^{1/2+\varepsilon} N^\varepsilon$ and likewise for v , and Proposition 8.1 yields

$$\sum_{k=1}^m \left| \left\langle \sum_{n \sim X} u_n e(\alpha n), \psi_k \right\rangle \overline{\left\langle \sum_{n \sim X} v_{n+h} e(\alpha n), \psi_k \right\rangle} \right| \ll_\varepsilon \left(\frac{X}{H} \right)^{1/2} D^{1/2+\varepsilon} N^{1/2+\varepsilon}. \quad (8.9)$$

With $DN \asymp X$, the right-hand side is $\ll_\varepsilon X^{1+\varepsilon} (H/X)^{1/2}$, i.e. a *square-root short-shift saving* of size $(H/X)^{1/2}$ relative to the $X^{1+\varepsilon}$ scale of Type-II energy.

Remarks. (i) The form (8.9) is stated abstractly so it can be plugged into any bilinear decomposition (Vaughan/Heath-Brown); only ℓ^2 -sizes of the bilinear pieces are used. (ii) No cancellation is assumed in u, v beyond divisor-type growth; the saving is *purely geometric* (a consequence of bank bandwidth). (iii) The bound is amplitude-invariant: scaling the test packets or the sequences scales both sides accordingly.

Uniform packet derivatives for far-band IBP

Let $\psi_{a/q}(\alpha)$ be a balanced packet (§4) centred at $\alpha = a/q$ with $q \leq Q$, width $\asymp 1/H$, built from a fixed even bump $w_0 \in C_c^\infty([-2, 2])$ via $W(\alpha) = w_0(H(\alpha - \alpha_{a/q}))$ and the rank-two projector P .

Lemma 8.2 (Level-uniform derivative bounds). *For every integer $c \geq 0$ there exists $C_c = C_c(w_0)$ such that, uniformly for all reduced centres a/q with $q \leq Q$,*

$$\sup_{\alpha \in \mathbf{TT}} |\partial_\alpha^c \psi_{a/q}(\alpha)| \leq C_c H^c, \quad \int_{-1/2}^{1/2} |\partial_\alpha^c \psi_{a/q}(\alpha)| d\alpha \leq C_c H^{c-1}.$$

Consequently, c integrations by parts against $e(\alpha t)$ yield a far-band decay factor $|t|^{-c}$ with constants independent of $q \leq Q$.

Proof. Write $\psi = PW$ with $P = I - \Pi$, Π the fixed rank-two projector from (??). Since $\partial_\alpha^c W(\alpha) = H^c w_0^{(c)}(H(\alpha - \alpha_{a/q}))$, both the sup-norm and the L^1 -norm scale by H^c and H^{c-1} , respectively, with constants depending only on w_0 and c . The same bounds hold for $\partial_\alpha^c \Pi W$ because Π is a fixed finite-rank integral operator with smooth kernel (moments against $1, \alpha$). Subtracting gives the claim. $\square$

Remark 8.3 (What is (not) used). No automorphic kernel enters here; the IBP is purely on $\mathbf{TT}$ for the additive packets. Thus there is no dependence on Eisenstein/oldforms, and the constants depend only on w_0 (and c), not on q .

Lemma 8.4 (Bank-aggregated short-shift bound). *Let $u(\alpha) = \sum_{n \in I} u_n e(\alpha n)$ and $v(\alpha) = \sum_{n \in I} v_n e(\alpha n)$ with $|I| = X$, and let $T_h v(\alpha) := e(h\alpha) v(\alpha)$ denote the shift by $h \in \mathbb{Z}$. With the same bank $\{\psi_k\}$ as in Proposition 9.4, one has for every h and every $H \geq 1$,*

$$\sum_k |\langle u, \psi_k \rangle \langle T_h v, \psi_k \rangle| \ll \left(\frac{X + Q^2}{H} \right)^{1/2} \left(\sum_{n \in I} |u_n|^2 \right)^{1/2} \left(\sum_{n \in I} |v_n|^2 \right)^{1/2}. \quad (8.10)$$

In particular, when $Q^2 \leq X$ this reads $\ll (X/H)^{1/2} \|u\|_2 \|v\|_2$.

Proof. By Cauchy–Schwarz over k ,

$$\sum_k |\langle u, \psi_k \rangle \langle T_h v, \psi_k \rangle| \leq \left(\sum_k |\langle u, \psi_k \rangle|^2 \right)^{1/2} \left(\sum_k |\langle T_h v, \psi_k \rangle|^2 \right)^{1/2}.$$

Apply Proposition 9.4 separately to u and to $T_h v$ (note T_h only modulates v and does not change the coefficient length X):

$$\sum_k |\langle u, \psi_k \rangle|^2 \ll \frac{X + Q^2}{H} \sum_{n \in I} |u_n|^2, \quad \sum_k |\langle T_h v, \psi_k \rangle|^2 \ll \frac{X + Q^2}{H} \sum_{n \in I} |v_n|^2.$$

Taking square roots and multiplying yields (8.10). $\square$

9 Type–I leakage: banked, amplitude–invariant ledger

9.1 Twin Type–I coefficients satisfy the Goldbach ledger hypotheses

Let $B(\alpha) = \sum_{n \sim X} b_n e(\alpha n)$ be a Type–I component from a Vaughan/Heath–Brown decomposition of $\Lambda(n)\Lambda(n+2)$. Then

$$b_n = \sum_{\substack{d|n \\ d \sim D}} a_d + \sum_{\substack{d|n+2 \\ d \sim D}} a'_d,$$

with $X^\varepsilon \leq D \leq X^{1-\varepsilon}$ and divisor–type bounds $|a_d|, |a'_d| \ll_\varepsilon d^\varepsilon$.

Lemma 9.1 (Twin Type–I ℓ^2 bound). *We have $\sum_{n \sim X} |b_n|^2 \ll_\varepsilon X^{1+\varepsilon}$ uniformly.*

Proof. Apply Cauchy–Schwarz and $\sum_{n \sim X} (\sum_{d|n, d \sim D} d^\varepsilon)^2 \ll X^{1+\varepsilon}$; the shift $n \mapsto n+2$ is harmless. $\square$

Lemma 9.2 (Twin Type–I ledger at constant–one scale). *With balanced packets ψ_k (two–moment projector on both legs), the Goldbach Type–I ledger carries over verbatim:*

$$\frac{1}{m} \sum_{k=1}^m |\langle B, \psi_k \rangle|^2 \ll \frac{X}{H Q^2} \sum_{n \sim X} |b_n|^2,$$

and the AO–refined bound for a subfamily of size L is

$$\sum_{k \in S} |\langle B, \psi_k \rangle|^2 \ll \left(\frac{L}{m} + \frac{\varepsilon}{1-\varepsilon} \right) \frac{X}{H Q^2} \sum_{n \sim X} |b_n|^2.$$

Proof. Identical to the Goldbach proof: the local additive large sieve with $q/\varphi(q)$ weights is independent of the arithmetic content of b_n (only ℓ^2 appears), plumbing to primitive characters is identical, and K1 kills the constant/alias block on both legs. Use Lemma 9.1. $\square$

Let $B(\alpha) = \sum_{n \sim X} b_n e(\alpha n)$ with coefficients (b_n) supported on $[X, 2X]$ and of divisor–type size $|b_n| \ll_\varepsilon n^\varepsilon$ (the Type–I channel after a Vaughan/Heath–Brown decomposition). Let $\{\psi_k\}_{k=1}^m$ be the balanced packets from §4, centered at reduced fractions a_k/q_k with $q_k \leq Q$ and width $\asymp 1/H$.

9.2 Local additive large sieve for balanced packets

Lemma 9.3 (Local LS at constant-one scale). *There exists a constant $C(w_0)$ depending only on the fixed bump w_0 and the bank overlap such that*

$$\sum_{k=1}^m |\langle B, \psi_k \rangle|^2 \leq C(w_0) \frac{X + Q^2}{H} \sum_{n \sim X} |b_n|^2. \quad (9.1)$$

The inequality is amplitude-invariant (homogeneous of degree 2 in B) and uniform in the placement of centers subject to the separation $\gg 1/H$ and the bound $q_k \leq Q$.

Proof. Write $\psi_k = (I - \Pi)W_k$ with $W_k(\alpha) = w_0(H(\alpha - \alpha_k))$ and Π the rank-two projector from §4 (so $\langle \psi_k, 1 \rangle = \langle \psi_k, \alpha \rangle = 0$ by (4.1)–(4.2)). Since Π is self-adjoint,

$$\langle B, \psi_k \rangle = \langle (I - \Pi)B, W_k \rangle,$$

hence by Cauchy–Schwarz,

$$\sum_k |\langle B, \psi_k \rangle|^2 \leq \sum_k |\langle B, W_k \rangle|^2 = \sum_k \left| \int_{\mathbb{T}} B(\alpha) w_0(H(\alpha - \alpha_k)) d\alpha \right|^2.$$

Let $w(\alpha) = w_0(H\alpha)$, so $\int_{\mathbb{T}} w(\alpha) d\alpha = H^{-1} \int w_0 = H^{-1}$. Interchanging the k -sum and the α -integral and expanding $|B|^2$ gives

$$\sum_k |\langle B, W_k \rangle|^2 = \sum_k \int_{\mathbb{T}} \left| \sum_{n \sim X} b_n e(\alpha n) \right|^2 w(\alpha - \alpha_k) d\alpha = \int_{\mathbb{T}} \left| \sum_{n \sim X} b_n e(\alpha n) \right|^2 \left(\sum_k w(\alpha - \alpha_k) \right) d\alpha.$$

The centers $\{\alpha_k\}$ are δ -separated with $\delta \asymp Q^{-2}$ (Farey geometry) and the translates of w have bounded overlap. Applying the *weighted additive large sieve* (Montgomery–Vaughan; see also [IK04, §13.3]) to the δ -separated family $\{\alpha_k\}$ with the nonnegative weight w (of total mass H^{-1}) yields

$$\sum_k \int_{\mathbb{T}} \left| \sum_{n \sim X} b_n e(\alpha n) \right|^2 w(\alpha - \alpha_k) d\alpha \ll \frac{X + \delta^{-1}}{H} \sum_{n \sim X} |b_n|^2 \ll \frac{X + Q^2}{H} \sum_{n \sim X} |b_n|^2.$$

Combining with the first inequality proves (9.1). The bound is homogeneous of degree 2 in B , hence amplitude-invariant, and the constants depend only on w_0 and the fixed overlap of the bank. $\square$

9.3 Averaging, AO dispersion, and the ledger line

Let m be the number of active packets. In our bank (one packet per reduced a/q with $q \leq Q$) we have $m \asymp \sum_{q \leq Q} \varphi(q) \asymp Q^2$.

Proposition 9.4 (Type-I ledger, bank-averaged). *With notation as above,*

$$\frac{1}{m} \sum_{k=1}^m |\langle B, \psi_k \rangle|^2 \ll \left(\frac{X}{H Q^2} + \frac{1}{H} \right) \sum_{n \sim X} |b_n|^2. \quad (9.2)$$

Moreover, for any subfamily $S \subset \{1, \dots, m\}$ of size L , the AO inequality (6.5) gives

$$\sum_{k \in S} |\langle B, \psi_k \rangle|^2 \leq \left(\frac{L}{m} + \frac{\varepsilon}{1 - \varepsilon} \right) \sum_{k=1}^m |\langle B, \psi_k \rangle|^2, \quad (9.3)$$

so the contribution of any fixed block of size L is bounded by

$$\ll \left(\frac{L}{m} + \frac{\varepsilon}{1 - \varepsilon} \right) \left(\frac{X}{H} + \frac{Q^2}{H} \right) \sum_{n \sim X} |b_n|^2.$$

Proof. Divide (9.2) by $m \asymp Q^2$ to get (9.2). Apply Proposition 6.2 to get (9.3). $\square$

Exact removal of the $1/H$ block by K1. The Q^2/H part in (9.2) is a finite-rank contribution supported on the span of $\{1, \alpha\}$ (constant and first alias). By Lemma 4.1, replacing raw windows by balanced packets on *both legs* kills this piece *exactly*. Hence the ledger for the balanced Type-I channel reads

$$\frac{1}{m} \sum_{k=1}^m |\langle B, \psi_k \rangle|^2 \ll \frac{X}{H Q^2} \sum_{n \sim X} |b_n|^2, \quad (9.4)$$

and the AO-refined bound for any subfamily S is

$$\sum_{k \in S} |\langle B, \psi_k \rangle|^2 \ll \left(\frac{L}{m} + \frac{\varepsilon}{1 - \varepsilon} \right) \frac{X}{H} \frac{1}{Q^2} \sum_{n \sim X} |b_n|^2. \quad (9.5)$$

All bounds are amplitude-invariant and independent of h (no shift appears in Type-I).

10 PSB/assembly: nonnegative extractor and twin pin

We now convert bank energy into a localized, nonnegative twin sum. The construction is completely amplitude-invariant and uses only positive-definite kernels.

10.1 Bank energy and a positive-definite weight

Define the *bank weight*

$$W_{\text{bank}}(\alpha) := \sum_{k=1}^m |\psi_k(\alpha)|^2 \geq 0.$$

By bounded overlap and packet width $\asymp 1/H$,

$$\int_{-1/2}^{1/2} W_{\text{bank}}(\alpha) d\alpha \asymp \frac{m}{H} \asymp \frac{Q^2}{H}, \quad \widehat{W}_{\text{bank}}(t) := \int_{-1/2}^{1/2} W_{\text{bank}}(\alpha) e(-\alpha t) d\alpha \quad (10.1)$$

satisfies $|\widehat{W}_{\text{bank}}(t)| \ll_A H^{-1} (1 + |t|/H)^{-A}$ for every fixed A .

For any finitely supported sequence $a = (a_n)$ with transform $A(\alpha) = \sum_n a_n e(\alpha n)$, Parseval gives the positive identity

$$\int_{-1/2}^{1/2} |A(\alpha)|^2 W_{\text{bank}}(\alpha) d\alpha = \sum_{t \in \mathbb{Z}} \widehat{W}_{\text{bank}}(t) \sum_n a_n \overline{a_{n-t}}. \quad (10.2)$$

10.2 Extractor J_H and normalization

Define the *bank extractor* $J_H : \mathbb{Z} \rightarrow [0, \infty)$ by

$$J_H(t) := \frac{\widehat{W}_{\text{bank}}(t)}{\widehat{W}_{\text{bank}}(0)}, \quad \text{so that } J_H(0) = 1, \quad J_H(t) \ll_A (1 + |t|/H)^{-A}. \quad (10.3)$$

Then J_H is positive-definite (Fourier transform of the nonnegative W_{bank}), nonnegative, and has effective bandwidth $\asymp H$. For sequences supported on $[X, 2X]$ we use the smoothed twin sum

$$\mathcal{C}_2(X; J_H) := \sum_{n \sim X} \Lambda(n) \Lambda(n+2) J_H(n-n) = \sum_{n \sim X} \Lambda(n) \Lambda(n+2) J_H(2),$$

or, more generally, with a mild n/X dependence one can replace $J_H(2)$ by a smooth $J_H(n/X)$ defined via a partition of unity; we keep the discrete normalization (10.3) for auditability.

10.3 Assembly inequality (PSB)

Apply (10.2) to $a_n = \Lambda(n)$ and extract the $t = 2$ contribution:

$$\int |S(\alpha)|^2 W_{\text{bank}}(\alpha) d\alpha \geq \widehat{W}_{\text{bank}}(2) \sum_{n \sim X} \Lambda(n) \Lambda(n+2), \quad (10.4)$$

since all terms are real and $\widehat{W}_{\text{bank}}(t) \geq 0$. Dividing by $\widehat{W}_{\text{bank}}(0)$ and using (10.3),

$$\sum_{n \sim X} \Lambda(n) \Lambda(n+2) J_H(2) \leq \frac{\int |S(\alpha)|^2 W_{\text{bank}}(\alpha) d\alpha}{\widehat{W}_{\text{bank}}(0)}. \quad (10.5)$$

More generally, introducing a standard dyadic partition in n and averaging (10.5) over short intervals gives the mesoscopic form with a *nonnegative* localized extractor $J_H(n/X)$ (details omitted; the discrete version suffices for the ledger).

10.4 From bank energy to localized twins

By Cauchy–Schwarz and the definition of W_{bank} ,

$$\int |S|^2 W_{\text{bank}} \geq \sum_{k=1}^m |\langle S, \psi_k \rangle|^2. \quad (10.6)$$

Combining (10.5) and (10.6) yields the *PSB assembly inequality*

$$\sum_{n \sim X} \Lambda(n) \Lambda(n+2) J_H(2) \leq \frac{1}{\widehat{W}_{\text{bank}}(0)} \sum_{k=1}^m |\langle S, \psi_k \rangle|^2. \quad (10.7)$$

Here $\widehat{W}_{\text{bank}}(0) \asymp Q^2/H$ by (10.1). The right-hand side is controlled by the ledger (Type–I via (9.5); Type–II via §8.4), and AO further limits how much any subfamily can capture.

10.5 Pinned singular series and conclusion (spine form)

Let $\mathfrak{S}(2)$ denote the twin singular series. The major-arc pin (carried in §11) associates to each packet a local density weight whose bank average equals $\mathfrak{S}(2) + O(Q^{-1+\varepsilon})$; since J_H is nonnegative and depends only on W_{bank} , the PSB inequality (10.7) together with the Type–I/II ledger bounds and §6.2 yields the *bank-local* positivity statement announced in §1.1, provided the falsifier gates pass with the frozen constants. All steps are amplitude–invariant, and all constants depend only on (ε, A, γ) and the fixed packet shape.

11 AR — Arithmetic routing: pin, SW window, plumbing, prime powers

Throughout we keep the frozen parameters of §1.1: $H = (\log X)^A$ with $A \geq 10$, $Q = H^\gamma$ with $0 < \gamma < \frac{1}{2}$, dyadics D, N obey $X^\varepsilon \leq D, N \leq X^{1-\varepsilon}$ and $DN \asymp X$. All statements are amplitude–invariant.

11.1 Pinned twin singular series (bank average)

Let $\mathfrak{S}(2)$ denote the twin singular series,

$$\mathfrak{S}(2) := 2 \prod_{p \geq 3} \frac{p(p-2)}{(p-1)^2}.$$

For a reduced center $\alpha = a/q$ ($q \leq Q$) write the local twin weight

$$\sigma_q(a; 2) := \#\{n \bmod q : n \equiv -a^{-1} \pmod{q}, n+2 \equiv -a^{-1} \pmod{q} \text{ mod prime powers dividing } q\},$$

normalized so that $0 \leq \sigma_q(a; 2) \leq 1$. The bank average of local densities reproduces $\mathfrak{S}(2)$ up to the natural Q^{-1} scale.

Lemma 11.1 (Pinned singular series, bank average). *Let $\mathcal{B}_Q = \{(a, q) : 1 \leq q \leq Q, (a, q) = 1\}$, and let $\nu_q := \varphi(q)$ be the multiplicity of centers at level q . Then*

$$\frac{1}{\sum_{q \leq Q} \nu_q} \sum_{q \leq Q} \sum_{\substack{a \bmod q \\ (a, q) = 1}} \sigma_q(a; 2) = \mathfrak{S}(2) + O_\varepsilon(Q^{-1+\varepsilon}). \quad (11.1)$$

The $O_\varepsilon(\cdot)$ is absolute once the normalization of σ_q is frozen. The bound is uniform in $Q \leq (\log X)^{A\gamma}$.

Sketch. By multiplicativity of local constraints and orthogonality of characters modulo prime powers, the average of $\sigma_q(a; 2)$ over $(a, q) = 1$ equals the Euler product truncated at $p \mid q$, $\prod_{p \nmid q} \frac{p(p-2)}{(p-1)^2}$. Averaging $q \leq Q$ fills in missing primes $p > Q$ up to error $\ll \sum_{p > Q} p^{-2} \ll Q^{-1}$, and smoothing across prime powers contributes at most $Q^{-1+\varepsilon}$ by Rankin–Selberg tricks. Details are routine and omitted. $\square$

11.2 Siegel–Walfisz window at polylog moduli

We quote a standard SW-type estimate uniform for $q \leq (\log X)^B$ with fixed B . For $\chi \bmod q$ and a smooth w supported on $[1, 2]$ with bounded derivatives,

$$\sum_n \Lambda(n) \chi(n) w\left(\frac{n}{X}\right) = \delta_\chi \widehat{w}(1) X + O_{B,w}\left(X \exp(-c \sqrt{\log X})\right), \quad (11.2)$$

where $\delta_\chi = 1$ only for the principal character, and $c = c(B) > 0$ is absolute. In particular, for $q \leq Q = (\log X)^{\gamma A}$ with $\gamma < \frac{1}{2}$, (11.2) holds uniformly with $B = \gamma A$.

Remark 11.2. No Siegel zero enters at the polylog scale. The bound (11.2) feeds all places we appeal to a Q -range prime distribution (major-arc pins and minor smoothing). We keep (11.2) segregated from the amplitude-invariant ledger so as not to overstate its role.

11.3 Imprimitive $\leftrightarrow$ primitive plumbing (constant-one scale)

Let $\mathcal{F}_Q$ denote the family of primitive characters with conductor $\leq Q$. For any coefficients (c_n) ,

$$\sum_{q \leq Q} \frac{q}{\varphi(q)} \sum_{\chi \bmod q}^* \left| \sum_{n \sim X} c_n \chi(n) \right|^2 = \sum_{\substack{\chi \text{ primitive} \\ \text{cond}(\chi) \leq Q}} \sum_{\chi \bmod q} \frac{q}{\varphi(q)} \left| \sum_{n \sim X} c_n \chi(n) \right|^2, \quad (11.3)$$

and the right-hand side equals

$$\sum_{\chi \in \mathcal{F}_Q} \left(\sum_{\substack{q \leq Q \\ \text{cond}(\chi) | q}} \frac{q}{\varphi(q)} \right) \left| \sum_{n \sim X} c_n \chi(n) \right|^2 \asymp \sum_{\chi \in \mathcal{F}_Q} \left| \sum_{n \sim X} c_n \chi(n) \right|^2, \quad (11.4)$$

since $\sum_{q \leq Q, q \equiv 0 \pmod{\text{cond}(\chi)} \frac{q}{\varphi(q)} \asymp 1$ with absolute constants. Thus the large sieve at constant-one scale,

$$\sum_{q \leq Q} \frac{q}{\varphi(q)} \sum_{\chi \bmod q}^* \left| \sum_{n \sim X} c_n \chi(n) \right|^2 \leq (X + Q^2) \sum_{n \sim X} |c_n|^2, \quad (11.5)$$

is *equivalent* to the same inequality with the sum restricted to $\mathcal{F}_Q$, up to absolute constants. No hidden q^ε factors appear.

11.4 Prime-power disposal for twins

Let $\Lambda_{\geq 2}(n) := \sum_{k \geq 2} \sum_p (\log p) \mathbf{1}_{n=p^k}$ be the prime-power part. Then

$$\sum_{n \sim X} \Lambda_{\geq 2}(n) \Lambda(n+2) + \sum_{n \sim X} \Lambda(n) \Lambda_{\geq 2}(n+2) \ll X^{1/2} (\log X)^2, \quad (11.6)$$

and, more generally, with any bounded nonnegative weight $W(n/X)$ of bandwidth $\asymp 1/H$,

$$\sum_{n \sim X} \Lambda_{\geq 2}(n) \Lambda(n+2) W(n/X) \ll X^{1/2} (\log X)^2. \quad (11.7)$$

Proof sketch. $\sum_{p^k \sim X, k \geq 2} \log p \ll X^{1/2} \log X$; pair trivially with $\Lambda(n+2) \ll \log X$ (or with Cauchy-Schwarz and $\sum \Lambda^2 \ll X \log X$) to get (11.6)–(11.7). $\square$

11.5 Vaughan/Heath-Brown interface and ℓ^2 sizes

We record the standard bilinear decomposition shapes and ℓ^2 -sizes used when invoking the SSU bound (§8.4) and the Type-I ledger (§9). For a smooth weight w supported on $[1, 2]$,

$$\sum_n \Lambda(n) w\left(\frac{n}{X}\right) = \mathbf{I}(X) + \mathbf{II}(X),$$

where $\mathbf{I}$ is a sum of Type-I forms $\sum_{n \sim X} b_n$ with $b_n = \sum_{d|n, d \sim D} a_d$ and $D \leq X^{1/2}$ (or $D \leq X^{1-\varepsilon}$ by Heath-Brown), and $\mathbf{II}$ is a bilinear form

$$\sum_{d \sim D} \sum_{n \sim N} a_d b_n \mathbf{1}_{dn \sim X}, \quad X^\varepsilon \leq D, N \leq X^{1-\varepsilon}, \quad DN \asymp X.$$

Divisor-type coefficient bounds $|a_d| \ll_\varepsilon d^\varepsilon$, $|b_n| \ll_\varepsilon n^\varepsilon$ imply

$$\left\| \sum_{d \sim D} a_d \mathbf{1}_d \right\|_{\ell^2([X, 2X])} \ll_\varepsilon D^{1/2+\varepsilon} X^\varepsilon, \quad \|b\|_{\ell^2([N, 2N])} \ll_\varepsilon N^{1/2+\varepsilon}. \quad (11.8)$$

Therefore, the SSU estimate (8.9) yields the advertised $(H/X)^{1/2}$ saving at constant-one scale for Type-II, while the Type-I ledger (9.5) controls the complementary channel after K1 cancellation.

Summary of routing. Pinned majors contribute $\mathfrak{S}(2) + O(Q^{-1+\varepsilon})$ by (11.1). The SW window (11.2) ensures uniform AP control for $q \leq Q$. Plumbing allows us to restrict to primitive characters at constant-one scale (11.3)–(11.5). Prime-powers are negligible by (11.7). The VHB interface (11.8) supplies the ℓ^2 sizes needed for the SSU bound and Type-I ledger.

Pinned majors: uniform positivity with bank

Let W_{bank} be as in (10.1) and J_H as in (10.3). Then

$$\int_{-1/2}^{1/2} |S(\alpha)|^2 W_{\text{bank}}(\alpha) d\alpha = \widehat{W}_{\text{bank}}(0) \frac{\mathfrak{S}(2)}{(\log X)^2} H (1 + o(1)) + \mathcal{E}_{\text{maj}}(X; H, Q), \quad (11.9)$$

with $\widehat{W}_{\text{bank}}(0) \asymp Q^2/H$ and

$$\mathcal{E}_{\text{maj}}(X; H, Q) \ll H (\log X)^{-2-\delta}$$

for some fixed $\delta > 0$ (depending only on the bump shape and $\gamma < 1/2$). Consequently,

$$\sum_{n \sim X} \Lambda(n) \Lambda(n+2) J_H(2) \geq \mathfrak{S}(2) \frac{H}{(\log X)^2} (1 + o(1)).$$

Sketch. Restrict to the union of major arcs $|\alpha - a/q| \leq (HQ)^{-1}$ with $q \leq Q$. On each arc, expand $S(\alpha)$ via Vaughan/Heath–Brown and evaluate the linearized integral with the additive character $e((\alpha - a/q)n)$; apply Siegel–Walfisz in AP up to $q \leq Q = (\log X)^{A\gamma}$; the singular series factors supply $\mathfrak{S}(2)$ by Lemma 11.1. Stationary phase and the smooth packet give a local factor proportional to $H/(\log X)^2$. Summing arcs and using bounded overlap of W_{bank} yields (11.9). The error comes from the SW remainder, dyadic edges, and the bank cutoffs, each bounded by $H(\log X)^{-2-\delta}$. $\square$

12 Banked operator bound and cross-level smoothing (CLS)

Theorem 12.1 (Banked operator bound from the two levers). *Assume Lever 1 (Cor. 5.8) and Lever 2 (Prop. 7.5). Then*

$$\|G\|_{\ell^2 \rightarrow \ell^2} \ll x^{o(1)} (H + Q^2) \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right).$$

Consequently, for the Type-II minor-arc energy,

$$\int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) d\alpha \ll H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2,$$

and, in the optional exponentiated packaging using good-bank dominance, $\|G\| \ll x^{o(1)} (H + Q^2)^{1-2\eta}$ with $\eta = \sigma_S/A + \gamma\delta_D + \zeta/(2A)$.

Proof. By Cor. 5.8, each diagonal satisfies $G_{q,q} \ll x^{o(1)} (H + Q^2) (\log x)^{-\sigma_S}$. By Prop. 7.5, for each row we have $\sum_{q_2 \neq q_1} |G_{q_1, q_2}| \ll (H + Q^2) Q^{-\delta_D + o(1)}$. Schur's test gives the operator bound. The minor-arc inequality follows from the TT* reduction with the usual normalization; the optional exponentiated form is the standard good-bank dominance packaging. $\square$

Our packaging mirrors the spectral-large-sieve paradigm (cf. [IK04, Ch. 16]).

We now combine the diagonal (§ 5) and off-diagonal (§ 7) inputs via Schur/Cotlar to control the banked Gram operator.

12.1 Banked operator bound and CLS-smooth corollary

Let $G = (G_{q_1, q_2})_{q_i \leq Q}$ be the banked Gram matrix on $I = [x, x + H]$ produced by the Type II TT* ledger, with $H = (\log x)^A$ and $Q = (\log x)^{A\gamma}$. Recall from S.3 that

$$\max_{q \leq Q} |G_{q, q}| \ll x^{o(1)} (H + Q^2) (\log x)^{-\sigma_S} \quad (\text{Corollary 5.8}), \quad (12.1)$$

and from D that

$$\sup_{q_1 \leq Q} \sum_{q_2 \neq q_1} \frac{|G_{q_1, q_2}|}{H + Q^2} \ll Q^{-\delta_D} \quad (\text{Proposition 7.5}). \quad (12.2)$$

Theorem 12.2 (Operator bound — soft form). *Under (12.1) and (12.2),*

$$\|G\|_{\ell^2 \rightarrow \ell^2} \ll x^{o(1)} (H + Q^2) \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right). \quad (12.3)$$

Equivalently, for any unit vector $\alpha = (\alpha_q)_{q \leq Q}$, $\sum_{q_1, q_2} \alpha_{q_1} G_{q_1, q_2} \overline{\alpha_{q_2}} \ll x^{o(1)} (H + Q^2) ((\log x)^{-\sigma_S} + Q^{-\delta_D})$. (All implied constants depend only on (ε, A, γ) and the fixed packet/extractor shapes.)

Proof. Write $G = D + O$ with D diagonal, O off-diagonal. By (12.1), $\|D\| \leq \max_{q \leq Q} |G_{q, q}| \ll x^{o(1)} (H + Q^2) (\log x)^{-\sigma_S}$. By (12.2) and Schur's test (or Cotlar), $\|O\| \ll x^{o(1)} (H + Q^2) Q^{-\delta_D}$. Sum the two contributions. $\square$

Remark 12.3 (Summary). The diagonal entries shrink by a fixed log-power (S.3), while the off-diagonal “cross-talk” across levels decays by a fixed power of Q (D.1–D.3). The operator norm is controlled by the larger of these two small numbers; hence the two-term bound (12.3).

Optional strengthened packaging. If you also apply the TI good-bank projection (augmented-span removal) with saving $(\log x)^{-\zeta}$ on any exceptional mass, then one may rewrite (12.3) in the exponentiated form

$$\|G\| \ll x^{o(1)} (H + Q^2)^{1-2\eta}, \quad \eta := \frac{\sigma_S}{A} + \gamma \delta_D + \frac{\zeta}{2A}, \quad (12.4)$$

using $(\log x)^{-\alpha} = H^{-\alpha/A}$ and $Q^{-\delta_D} = (H + Q^2)^{-\gamma \delta_D + o(1)}$. This is purely cosmetic but convenient for the minor-arc corollary below. (If you prefer not to rely on good-bank projection here, keep the soft two-term form (12.3).)

Corollary 12.4 (CLS-smooth, Type II — soft form). *On the minor arcs $\mathfrak{m}$,*

$$\int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) d\alpha \ll H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2, \quad (12.5)$$

where $S(\alpha)$ is the Type II exponential sum for the twin payload over $I = [x, x + H]$.

Proof. Dualize the banked inequality underlying $\|G\|$ to the family $\{K_q\}_{q \leq Q}$ restricted to $\mathfrak{m}$, as in the TT* argument; the diagonal/off-diagonal control from (12.3) yields the stated two-term bound for the quadratic form $\int_{\mathfrak{m}} |S|^2 e(-2\alpha)$. $\square$

Corollary 12.5 (CLS-smooth, Type II — exponentiated form). *If you adopt (12.4), then*

$$\int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) d\alpha \ll H \left(\frac{Q}{H^\eta} \right)^2 = \frac{H}{(\log x)^{2A(\eta - \gamma)}}, \quad (12.6)$$

and in particular this is $\ll H/(\log x)^{2+\varepsilon_0}$ once $2A(\eta - \gamma) > 1$ for some $\varepsilon_0 > 0$.

Remark 12.6 (How to read the inequality). In either form, we get a quantitative saving on the minor arcs that beats the parity barrier at the polylogarithmic scale once the (mild) parameter inequality is met. The soft form (12.5) is unconditional under S.3 and D; the exponentiated form (12.6) simply packages the same savings as a single H -exponent (and optionally folds in the good-bank ζ).

Let $G = (G_{q_1, q_2})_{q_i \leq Q}$ be the banked Gram matrix on $I = [x, x + H]$.

Theorem 12.7 (WP3.5). *There exists $\eta > 0$ such that*

$$\|G\|_{\ell^2 \rightarrow \ell^2} \ll x^{o(1)} (H + Q^2)^{1-2\eta}, \quad \eta = \frac{\sigma_S}{A} + \gamma \delta_D + \frac{\zeta}{2A}.$$

(All implied constants depend only on (ε, A, γ) and the fixed packet/extractor shapes.)

Proof. Write $G = D + O$ with D diagonal and O off-diagonal. By Corollary 5.8, $\|D\| \leq \max_{q \leq Q} |G_{q,q}| \ll x^{o(1)} (H + Q^2) (\log x)^{-\sigma_S}$. By Proposition 7.5 and Schur's test, $\|O\| \ll x^{o(1)} (H + Q^2) Q^{-\delta_D}$. Project to the good bank (vector large sieve / augmented-span removal) to gain an extra factor $(\log x)^{-\zeta}$ on any exceptional mass. Rewrite $(\log x)^{-\alpha} = H^{-\alpha/A}$ and $Q^{-\delta_D} = (H + Q^2)^{-\gamma \delta_D + o(1)}$. Absorb the three savings into a single exponent 2η to obtain the display. $\square$

Corollary 12.8 (CLS-smooth, Type II). *On the minor arcs,*

$$\int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) d\alpha \ll H \left(\frac{Q}{H^\eta} \right)^2 = \frac{H}{(\log x)^{2A(\eta-\gamma)}}.$$

If $2A(\eta - \gamma) > 1$, this is $\ll H/(\log x)^{2+\varepsilon_0}$ for some $\varepsilon_0 > 0$.

13 CLS-smooth (Type II)

13.1 Banked minor arcs and the TT* reduction

Let $I = [x, x + H]$, $H = (\log x)^A$, $Q = (\log x)^{A\gamma}$, and let $\mathfrak{M}$ denote the AO bank of major arcs of width $\asymp \mu := Q^2/H$ around rationals a/q with $q \leq Q$. Set $\mathfrak{m} := \mathbb{T} \setminus \mathfrak{M}$.

Write $S(\alpha)$ for the (smoothed) Type II exponential sum coding the twin payload on I . As in the TT* ledger, after bank localization we may write

$$S(\alpha) = \sum_{q \leq Q} F_q(\alpha), \quad F_q := \text{the } q\text{-packet contribution (supported on } \mathfrak{M} \text{ but analyzed on all } \mathbb{T}).$$

Let $G = (G_{q_1, q_2})_{q_i \leq Q}$ be the banked Gram matrix:

$$G_{q_1, q_2} := \int_{\mathfrak{m}} F_{q_1}(\alpha) \overline{F_{q_2}(\alpha)} d\alpha.$$

Lemma 13.1 (Energy/operator equivalence on $\mathfrak{m}$). *For any complex vector $\beta = (\beta_q)_{q \leq Q}$,*

$$\int_{\mathfrak{m}} \left| \sum_{q \leq Q} \beta_q F_q(\alpha) \right|^2 d\alpha = \sum_{q_1, q_2} \beta_{q_1} G_{q_1, q_2} \overline{\beta_{q_2}} \leq \|G\|_{\ell^2 \rightarrow \ell^2} \sum_{q \leq Q} |\beta_q|^2.$$

In particular, with $\beta_q \equiv 1$ we have $\int_{\mathfrak{m}} |S(\alpha)|^2 d\alpha \leq \|G\| Q$.

The TT*-to-operator reduction is standard in this context; see [IK04, Ch. 16].

Proof. Expand the square and use Fubini; the inequality is the operator norm bound for G . $\square$

13.2 CLS-smooth: soft and exponentiated forms

Theorem 13.2 (CLS-smooth: soft form). *Assume the diagonal one-level saving and cross-level decay (Corollary 5.8 and Proposition 7.5). Then*

$$\int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) d\alpha \ll H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2.$$

Proof. By Cauchy–Schwarz, $\left| \int_{\mathfrak{m}} |S|^2 e(-2\alpha) d\alpha \right| \leq \int_{\mathfrak{m}} |S|^2 d\alpha$. Apply Lemma 13.1 with $\beta_q \equiv 1$: $\int_{\mathfrak{m}} |S|^2 \leq \|G\| Q$. By the operator bound,

$$\|G\| \ll x^{o(1)} (H + Q^2) \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right).$$

Since $Q \leq (H + Q^2)^{1/2+o(1)}$, the stated H -level inequality follows. $\square$

Corollary 13.3 (CLS-smooth: exponentiated packaging). *If, in addition, the good-bank dominance with exponent $\zeta > 0$ is applied so that $\|G\| \ll x^{o(1)} (H + Q^2)^{1-2\eta}$ with $\eta = \sigma_S/A + \gamma\delta_D + \zeta/(2A)$, then*

$$\int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) d\alpha \ll H \left(\frac{Q}{H^\eta} \right)^2 = \frac{H}{(\log x)^{2A(\eta-\gamma)}}.$$

13.3 Uniform-in- t CLS for the gate

Lemma 13.4 (Uniform shift). *For every integer t ,*

$$\left| \int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) e(\alpha t) d\alpha \right| \leq \int_{\mathfrak{m}} |S(\alpha)|^2 d\alpha \ll H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2,$$

and, under Corollary 13.3, also $\ll H/(\log x)^{2A(\eta-\gamma)}$.

Proof. Insert $e(\alpha t)$, take absolute values, and apply Theorem 13.2 (or Corollary 13.3). $\square$

13.4 Bank-scale smoothing (optional)

Let Υ be any even C_c^∞ bump on $\mathbb{T}$ with $\int \Upsilon = 1$, and write $\Upsilon_\mu(\alpha) = \mu^{-1} \Upsilon(\alpha/\mu)$ with $\mu \asymp Q^2/H$.

Lemma 13.5 (CLS with bank-scale smoothing). *For any $B \geq 0$,*

$$\int_{\mathfrak{m}} (|S|^2 * \Upsilon_\mu)(\alpha) e(-2\alpha) d\alpha \ll_B H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2,$$

and the same bound holds with a weight $(1 + |t|\mu)^B$ on the Fourier side.

Proof. Convolution with Υ_μ multiplies the Fourier coefficients by $\widehat{\Upsilon}(\mu t)$, which is $\ll_B (1 + |t|\mu)^{-B}$. Combine with Lemma 13.4 by Parseval. $\square$

From Theorem 12.7 we obtain

$$\int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) d\alpha \ll H \left(\frac{Q}{H^\eta} \right)^2 = \frac{H}{(\log x)^{2A(\eta-\gamma)}}.$$

Choose parameters with $2A(\eta - \gamma) > 1$; see §15.

Remark 13.6 (How CLS is used in the Gate). Expanding the pin $W(\alpha) = \sum_t K(t)e(-t\alpha)$ and applying Lemma 13.4, we obtain

$$\left| \int_{\mathfrak{m}} |S|^2 e(-2\alpha) W(\alpha) d\alpha \right| \leq \sum_t |K(t)| \cdot \int_{\mathfrak{m}} |S|^2 d\alpha \ll \left(\sum_t |K(t)| \right) \cdot H((\log x)^{-\sigma_S} + Q^{-\delta_D})^2.$$

By the pin's Fourier tail (constructed in the next section), $\sum_t |K(t)| \ll 1$, or with Lemma 13.5 we may work with the moment $\sum_t |K(t)|(1 + |t|\mu)^B \ll_B 1$.

Parameter inequality. In the exponentiated packaging, the CLS bound is $\ll H/(\log x)^{2A(\eta-\gamma)}$ with $\eta = \sigma_S/A + \gamma\delta_D + \zeta/(2A)$. Throughout we ensure $2A(\eta - \gamma) > 1$ by the explicit tuple recorded in the parameter audit.

14 Gate, pin, and pinned majors

14.1 Gate set-up and uniform minor-arc control

Let $I = [x, x + H]$, $H = (\log x)^A$, $Q = (\log x)^{A\gamma}$, and let $S(\alpha)$ denote the Type II exponential sum for the twin payload over I . Let $W \geq 0$ be the (to-be-specified) pin supported on the banked major arcs $\mathfrak{M}$ with $\int_{\mathbb{T}} W = 1$, and write its discrete Fourier series

$$W(\alpha) = \sum_{t \in \mathbb{Z}} K(t) e(-t\alpha), \quad K(t) = \int_{\mathbb{T}} W(\alpha) e(\alpha t) d\alpha, \quad (14.1)$$

so $K(0) = 1$. The gate identity splits the true-twin second moment as

$$\sum_{n \in I} \Lambda(n) \Lambda(n+2) = \int_{\mathfrak{M}} |S(\alpha)|^2 e(-2\alpha) W(\alpha) d\alpha \quad (14.2)$$

$$+ \int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) W(\alpha) d\alpha. \quad (14.3)$$

Lemma 14.1 (Uniform-in- t CLS-smooth). *For every $t \in \mathbb{Z}$,*

$$\left| \int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) e(\alpha t) d\alpha \right| \leq \int_{\mathfrak{m}} |S(\alpha)|^2 d\alpha \ll H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2, \quad (14.4)$$

and, under the exponentiated packaging,

$$\left| \int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) e(\alpha t) d\alpha \right| \ll \frac{H}{(\log x)^{2A(\eta-\gamma)}}. \quad (14.5)$$

Proof. Insert $e(\alpha t)$ and take absolute values; the triangle inequality gives $|\int_{\mathfrak{m}} |S|^2 e(-2\alpha) e(\alpha t)| \leq \int_{\mathfrak{m}} |S|^2$. Apply Corollary WP3.5/CLS in the soft (resp. exponentiated) form. $\square$

Proposition 14.2 (Gate reduction to a discrete tail). *With $K(t)$ as in (14.1),*

$$\left| \int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) W(\alpha) d\alpha \right| \leq \sum_{t \in \mathbb{Z}} |K(t)| \cdot \left| \int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) e(\alpha t) d\alpha \right|. \quad (14.6)$$

Consequently, by Lemma 14.1,

$$\left| \int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) W(\alpha) d\alpha \right| \ll \left(\sum_{t \in \mathbb{Z}} |K(t)| \right) \cdot H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2, \quad (14.7)$$

and similarly with the exponentiated CLS bound.

Remark 14.3 (What remains). To close the gate, we will build the pin W so that its Fourier tail satisfies

$$\sum_{t \in \mathbb{Z}} |K(t)| (1 + |t|\mu)^B \ll_B 1, \quad \mu \asymp \frac{Q^2}{H}, \quad (14.8)$$

with B large; and then we will control the unweighted $\sum_t |K(t)|$ by a standard B -to-0 interpolation (or pair it with a mild t -dependent smoothing of the CLS integral at the bank scale μ). Either route makes the RHS of (14.7) $\ll H/(\log x)^{2+\varepsilon_0}$, which is negligible against the pinned major-arc main term.

Remark 14.4 (Summary). This step shows two facts: (i) the CLS minor-arc saving is *uniform* in the Fourier shift t that appears when we expand the pin W ; (ii) the minor-arc piece equals a discrete convolution of the kernel tail $K(t)$ with those uniformly bounded integrals. The next step supplies a pin with a rapidly decaying tail, so the discrete convolution is tiny.

Let $W \geq 0$ be the banked twin pin supported on the packetized major arcs $\mathfrak{M}$, built from packet bumps $\Pi_{q,a}$ with weights $\mu_q(a)e(-2a/q)$ and normalized by $\int_{\mathbb{T}} W = 1$. Define $K(t) = \int_{\mathbb{T}} W(\alpha)e(\alpha t) d\alpha$.

Lemma 14.5 (Pin properties). $K(0) = 1$ and, for any $B \geq 1$,

$$\sum_{t \neq 0} |K(t)| (1 + |t|\mu)^B \ll_B 1, \quad \mu \asymp \frac{Q^2}{H}.$$

Moreover, W is orthogonal to the augmented local span at each $q \leq Q$.

Proof. W is a convex combination of smooth packet bumps, hence $W \geq 0$ and $\int W = 1$, giving $K(0) = 1$. Each packet bump has Fourier transform decaying faster than $(1 + |t|\text{width})^{-B}$ for any B . The packet width on the frequency side is $\asymp \mu = Q^2/H$, hence the stated tail bound after summing packets. Orthogonality to the augmented span holds by construction of the projector weights $\mu_q(a)$ (the local pretentious profiles and possible exceptional components lie in the span killed by the mean-zero projector). $\square$

Theorem 14.6 (Pinned majors). With W as above and $S(\alpha)$ the Type-II exponential sum for the twin payload,

$$\int_{\mathfrak{M}} S(\alpha) \overline{S(\alpha)} e(-2\alpha) W(\alpha) d\alpha = \mathfrak{S}_2 \frac{H}{(\log x)^2} (1 + o(1)),$$

uniformly in the Landau–Page dichotomy.

Proof. This is the standard major-arc evaluation (as in the Goldbach paper), with $e(-2a/q)$ encoded in the pin. The principal/principal contribution produces the product of local densities $\mathfrak{S}_2$ times $H/(\log x)^2$. Terms arising from characters/exceptional profiles lie in the augmented span and are annihilated by the projector; any residual mixed terms are bounded by $(\log x)^{-A}$ due to smooth Siegel–Walfisz on polylog moduli. $\square$

Theorem 14.7 (Gate inequality & pointwise lower bound). Let $H = (\log x)^A$. Then

$$\sum_{n \in [x, x+H]} \Lambda(n) \Lambda(n+2) \geq \mathfrak{S}_2 \frac{H}{(\log x)^2} - \left| \int_{\mathfrak{m}} S \overline{S} e(-2\alpha) W(\alpha) d\alpha \right|.$$

Moreover,

$$\left| \int_{\mathfrak{m}} S \overline{S} e(-2\alpha) W(\alpha) d\alpha \right| \leq \sum_{t \neq 0} |K(t)| \cdot \left| \int_{\mathfrak{m}} |S(\alpha)|^2 e(-2\alpha) e(\alpha t) d\alpha \right|.$$

By Lemma 14.5 and Corollary 12.8, the RHS is $\ll H/(\log x)^{2+\varepsilon_0}$; hence

$$\sum_{n \in [x, x+H]} \Lambda(n) \Lambda(n+2) \geq c(A) \frac{H}{(\log x)^2},$$

for some $c(A) > 0$ and all sufficiently large x .

Corollary 14.8 (Desmoothing and prime-power removal). *Replacing the smooth window by $1_{[x, x+H]}$ alters the sum by $O(H/(\log x)^{2+\varepsilon})$. Prime-power terms contribute $O((\log x)^{O(1)})$, negligible against $H/(\log x)^2$. Thus Theorem 14.7 follows.*

Uniformity. We only need log-free Siegel–Walfisz on moduli $q \leq (\log x)^B$ (for a fixed B); together with the augmented-span projector this removes pretentious characters uniformly across the Landau–Page dichotomy.

14.2 The pin W , pinned majors, and closing the gate

Recall $H = (\log x)^A$, $Q = (\log x)^{A\gamma}$, and let

$$\mu \asymp \frac{Q^2}{H} \quad (14.9)$$

be the bank frequency scale. Let $\{\Pi_{q,a}\}$ be a smooth partition of unity on the banked major arcs $\mathfrak{M}$, indexed by coprime fractions a/q with $q \leq Q$, such that each packet $\Pi_{q,a}$ is supported on

$$\left| \alpha - \frac{a}{q} \right| \ll \frac{\mu}{q^2}, \quad \int_{\mathbb{T}} \Pi_{q,a}(\alpha) d\alpha \asymp \frac{\mu}{q^2}, \quad \Pi_{q,a} \in C_c^\infty, \quad \|\partial_\alpha^r \Pi_{q,a}\|_\infty \ll_r \left(\frac{q^2}{\mu} \right)^r. \quad (14.10)$$

(Explicitly, take a fixed bump $\phi \in C_c^\infty([-2, 2])$ with $\phi \geq 0$, $\int \phi = 1$, and set $\Pi_{q,a}(\alpha) = C_q \phi\left(\frac{q^2}{A_0 \mu}(\alpha - \frac{a}{q})\right)$ with $C_q \asymp \mu/q^2$ and A_0 a large absolute constant.)

A. A phase-locked banked pin. Define the *phase-locked* pin

$$W(\alpha) := \frac{1}{Z} \sum_{q \leq Q} w_q \sum_{\substack{a \pmod{q} \\ (a(a+2), q)=1}} \Pi_{q,a}(\alpha) e\left(-\frac{2a}{q}\right), \quad Z = \int_{\mathbb{T}} \sum_{q \leq Q} w_q \sum_{(a(a+2), q)=1} \Pi_{q,a}(\alpha) d\alpha, \quad (14.11)$$

with weights $w_q \asymp 1$ smooth in q (e.g. $w_q = \varphi(q)$ or $w_q = 1$). Then W is supported on $\mathfrak{M}$ and $\int_{\mathbb{T}} W = 1$. Let $K(t)$ be its Fourier coefficients, as in (14.1).

Lemma 14.9 (Fourier tail of the pin). *For each $B \geq 1$,*

$$\sum_{t \in \mathbb{Z}} |K(t)| (1 + |t| \mu)^B \ll_B 1. \quad (14.12)$$

Consequently $\sum_t |K(t)| \ll 1$, and $|K(0)| = 1$.

Proof. By (14.10), each packet has Fourier transform $\widehat{\Pi}_{q,a}(t) = \int \Pi_{q,a}(\alpha) e(\alpha t) d\alpha$ satisfying, for all $B \geq 0$,

$$|\widehat{\Pi}_{q,a}(t)| \ll_B \frac{\mu}{q^2} (1 + |t| \mu/q^2)^{-B}. \quad (14.13)$$

Therefore $K(t) = Z^{-1} \sum_{q \leq Q} w_q \sum_{(a(a+2), q)=1} \widehat{\Pi}_{q,a}(t) e(-2a/q)$. Summing (14.13) over a gives $\ll \mu \sum_{q \leq Q} w_q q^{-2} (1 + |t| \mu/q^2)^{-B}$. A dyadic partition in q and the monotonicity of $u \mapsto (1 + |t| \mu/u)^{-B}$ yield $|K(t)| \ll_B (1 + |t| \mu)^{-B}$. Summing over t gives (14.12) (since $\sum_t (1 + |t| \mu)^{-B} \ll_B \mu^{-1}$ and $Z \asymp \mu \sum_{q \leq Q} w_q q^{-2} \asymp 1$), and $K(0) = 1$ follows from $\int W = 1$. $\square$

B. Pinned major arcs. Write the twin payload $S(\alpha)$ on $I = [x, x + H]$ in its standard factorized major-arc shape (Type II—with smooth window and with W -trick reinstated in the Euler factors). Then

$$\int_{\mathfrak{M}} |S(\alpha)|^2 e(-2\alpha) W(\alpha) d\alpha = \mathfrak{S}_2 \frac{H}{(\log x)^2} (1 + o(1)), \quad (14.14)$$

uniformly across the Landau–Page dichotomy, using log-free Siegel–Walfisz on moduli $\ll (\log x)^B$ and the augmented-span projector to annihilate pretentious pieces (exactly as in the Goldbach TI paper).

Proof. Write W via the packetized major arcs (8.11): for α near a/q with $q \leq Q$,

$$W(\alpha) = \frac{1}{Z} \sum_{q \leq Q} w_q \sum_{\substack{a \pmod{q} \\ (a(a+2), q) = 1}} \Pi_{q,a}(\alpha) e\left(-\frac{2a}{q}\right), \quad Z = \int_{\mathbb{T}} \sum_{q \leq Q} w_q \sum_{\substack{a \pmod{q} \\ (a(a+2), q) = 1}} \Pi_{q,a}(\alpha) d\alpha$$

with $\Pi_{q,a}$ satisfying the localization and smoothness bounds (8.10). Thus

$$\begin{aligned} \mathcal{I} &:= \int_{\mathcal{M}} S(\alpha) \overline{S(\alpha)} e(-2\alpha) W(\alpha) d\alpha \\ &= \frac{1}{Z} \sum_{q \leq Q} w_q \sum_{\substack{a \pmod{q} \\ (a(a+2), q) = 1}} e\left(-\frac{2a}{q}\right) \int_{\mathbb{T}} \Pi_{q,a}(\alpha) |S(\alpha)|^2 e(-2\alpha) d\alpha. \end{aligned}$$

Fix $q \leq Q$ and a with $(a(a+2), q) = 1$. On the support of $\Pi_{q,a}$ write $\alpha = \frac{a}{q} + \beta$ with $|\beta| \ll \mu/q^2$ and expand $S(\alpha)$ by orthogonality of Dirichlet characters:

$$S\left(\frac{a}{q} + \beta\right) = \frac{1}{\varphi(q)} \sum_{\chi \pmod{q}} \overline{\chi}(a) S_{\chi}(\beta) + \mathcal{E}_{q,a}(\beta), \quad S_{\chi}(\beta) := \sum_{n \in I} \Lambda(n) \chi(n) e(\beta n).$$

Here $\mathcal{E}_{q,a}$ absorbs the (negligible) contribution from $(n, q) > 1$; by the standard major-arc clean-up, $\int |\mathcal{E}_{q,a}|^2 \ll H(\log x)^{-A}$ uniformly for $q \leq (\log x)^B$ (log-free Siegel–Walfisz), see e.g. [MV07a, Ch. 13] and [IK04, §13.7]. Thus

$$|S|^2 = \frac{1}{\varphi(q)^2} \sum_{\chi_1, \chi_2 \pmod{q}} \overline{\chi_1}(a) \chi_2(a) S_{\chi_1}(\beta) \overline{S_{\chi_2}(\beta)} + O(H(\log x)^{-A}),$$

after integration against $\Pi_{q,a}(\alpha)$, uniformly in a .

Principal–principal block. For $\chi_1 = \chi_2 = \chi_0$ (principal character), by the major-arc model (PNT in short APs with smooth weight) we have, uniformly for $|\beta| \ll \mu/q^2$ and $q \leq (\log x)^B$,

$$S_{\chi_0}(\beta) = \sum_{\substack{n \in I \\ (n, q) = 1}} \Lambda(n) e(\beta n) = \frac{\varphi(q)}{q} \widehat{\mathbf{1}}_I(\beta) + O(H(\log x)^{-A}),$$

and in our Type II normalization this carries a factor $(\log x)^{-1}$ from each Λ (see the setup of §7 and the packaging preceding (8.4)–(8.5)), giving

$$S_{\chi_0}(\beta) = \frac{\varphi(q)}{q} \frac{\widehat{\mathbf{1}}_I(\beta)}{\log x} + O(H(\log x)^{-A}).$$

Consequently,

$$|S_{\chi_0}(\beta)|^2 = \left(\frac{\varphi(q)}{q} \right)^2 \frac{|\widehat{\mathbf{1}_I}(\beta)|^2}{(\log x)^2} + O(H^2(\log x)^{-1-A}).$$

Integrating against $\Pi_{q,a}(\frac{a}{q} + \beta) e(-2\beta)$ and using $\int |\widehat{\mathbf{1}_I}(\beta)|^2 \Pi_{q,a}(\frac{a}{q} + \beta) d\beta = (1+o(1)) \int_{\mathbb{R}} |\widehat{\mathbf{1}_I}(\beta)|^2 d\beta = (1+o(1)) H$ (Plancherel, packet width $\asymp \mu$), we obtain

$$\int \Pi_{q,a}(\alpha) |S(\alpha)|^2 e(-2\alpha) d\alpha = \frac{1}{\varphi(q)^2} |c_q(a)|^2 \frac{H}{(\log x)^2} e\left(-\frac{2a}{q}\right) + O\left(\frac{H}{(\log x)^{2+A}}\right),$$

where $c_q(a) = \sum_{r(\bmod q)}^* e(ar/q)$ is the Ramanujan sum, and $|c_q(a)| = |\mu(q)|$ for $(a, q) = 1$.

Mixed and nonprincipal blocks. Terms with at least one nonprincipal character are either (i) killed by the *augmented local span* orthogonality of W (Lemma 8.5), or (ii) bounded by $(\log x)^{-A}$ by log-free Siegel–Walfisz on moduli $\leq (\log x)^B$ (uniform in the Landau–Page dichotomy); see (8.5) and the discussion around (8.4)–(8.7). Hence their total contribution is $O(H(\log x)^{-2-A})$ after summing (q, a) .

Summing packets and extracting the singular series. Insert the principal–principal evaluation into the packet sum:

$$\mathcal{I} = \frac{H}{(\log x)^2} \frac{1}{Z} \sum_{q \leq Q} w_q \sum_{\substack{a \bmod q \\ (a(a+2), q)=1}} \frac{|c_q(a)|^2}{\varphi(q)^2} e\left(-\frac{2a}{q}\right) + O\left(\frac{H}{(\log x)^{2+A}}\right).$$

Because $|c_q(a)|^2 = \mu(q)^2$ for $(a, q) = 1$, and the inner sum equals $\mu(q)^2 c_q(2)$, we get

$$\mathcal{I} = \frac{H}{(\log x)^2} \frac{1}{Z} \sum_{q \leq Q} w_q \frac{\mu(q)^2 c_q(2)}{\varphi(q)^2} + O\left(\frac{H}{(\log x)^{2+A}}\right).$$

By the choice of w_q and the normalization $Z \asymp \sum_{q \leq Q} w_q \mu/q^2$ (see (8.11)–(8.13)), the truncated sum converges to the twin singular series

$$\mathfrak{S}_2 = \sum_{q=1}^{\infty} \frac{\mu(q)^2 c_q(2)}{\varphi(q)^2} = \prod_p \left(1 - \frac{1}{(p-1)^2}\right) \cdot \sigma_2(p),$$

with the usual local factor at $p = 2$, and the tail $q > Q$ is $o(1)$ because $Q = (\log x)^{A\gamma} \rightarrow \infty$ while packets have width $\mu \asymp Q^2/H$ (cf. (8.9)–(8.13)). Therefore

$$\mathcal{I} = \mathfrak{S}_2 \frac{H}{(\log x)^2} (1 + o(1)),$$

uniformly in the Landau–Page dichotomy, completing the proof. $\square$

The singular-series local factors and their stability under our pin are classical; cf. [MV07a, Ch. 1, 27] and [Kou19, Ch. 2].

C. Closing the gate. From (14.3) and Proposition 14.2,

$$\begin{aligned} \left| \int_{\mathfrak{m}} |S|^2 e(-2\alpha) W(\alpha) d\alpha \right| &\leq \sum_t |K(t)| \left| \int_{\mathfrak{m}} |S|^2 e(-2\alpha) e(\alpha t) d\alpha \right| \\ &\ll \left(\sum_t |K(t)| \right) \cdot H \left((\log x)^{-\sigma_S} + Q^{-\delta_D} \right)^2, \end{aligned}$$

by Lemma 14.1. By Lemma 14.9, $\sum_t |K(t)| \ll 1$, so the RHS is $o\left(\frac{H}{(\log x)^2}\right)$ as soon as either the soft form gives a fixed $\sigma_S > 0$ or the exponentiated form yields $2A(\eta - \gamma) > 1$. Combining (14.2), (14.14), and the bound above gives

$$\sum_{n \in [x, x+H]} \Lambda(n) \Lambda(n+2) \geq \mathfrak{S}_2 \frac{H}{(\log x)^2} - o\left(\frac{H}{(\log x)^2}\right) \geq c(A) \frac{H}{(\log x)^2}, \quad (14.15)$$

for some $c(A) > 0$ and all sufficiently large x . Desmoothing of the window and removal of prime-power terms are routine and change the sum by $O(H/(\log x)^{2+\varepsilon})$.

Remark 14.10 (Summary). W is a “phase-locked” spotlight placed on the major arcs: each packet around a/q carries the phase $e(-2a/q)$ so that the twin shift aligns. Its Fourier tail decays on the natural bank scale $\mu \asymp Q^2/H$, hence the minor-arc contribution—already small by CLS—is crushed after weighting by W . The major arcs deliver the expected main term with the twin singular series, giving a positive pointwise lower bound in every polylogarithmic window.

15 Parameters

Take, for instance, $A = 8$, $\gamma = 0.40$, $\sigma_S = 3.5$, $\delta_D = 0.05$, $\zeta = 0.10$. Then $2A(\eta - \gamma) = 1.02 > 1$, giving the needed margin in § 13.

15.1 Parameters and a worked tuple

We take the following.

$$H = (\log x)^A, \quad Q = (\log x)^{A\gamma},$$

and exponents $\sigma_S > 0$ (S.3 one-level log saving), $\delta_D > 0$ (cross-level decay from D.1–D.3), and $\zeta > 0$ (good-bank dominance). The WP3.5 exponent is

$$\eta = \frac{\sigma_S}{A} + \gamma \delta_D + \frac{\zeta}{2A},$$

and CLS-smooth yields a saving of size $(\log x)^{-2A(\eta - \gamma)}$ on the minor arcs. We require

$$2A(\eta - \gamma) > 1.$$

Example (admissible tuple). Take

$$A = 8, \quad \gamma = 0.40, \quad \sigma_S = 3.5, \quad \delta_D = 0.05, \quad \zeta = 0.10.$$

Then

$$\frac{\sigma_S}{A} = 0.4375, \quad \gamma \delta_D = 0.02, \quad \frac{\zeta}{2A} = 0.00625, \quad \Rightarrow \quad \eta = 0.46375,$$

and

$$2A(\eta - \gamma) = 16 \cdot (0.46375 - 0.40) = 16 \cdot 0.06375 = 1.02 > 1,$$

giving the needed margin in Cor. 12.8.

15.2 One explicit admissible tuple

Recall $\eta = \sigma_S/A + \gamma \delta_D + \zeta/(2A)$ and the requirement $2A(\eta - \gamma) > 1$. Take, for example,

$$(\sigma_S, \delta_D, \zeta) = (3, \tfrac{1}{2}, 2), \quad (A, \gamma) = (20, 0.30).$$

Then

$$\eta = \frac{3}{20} + 0.30 \cdot \frac{1}{2} + \frac{2}{40} = 0.35, \quad 2A(\eta - \gamma) = 40 \cdot (0.35 - 0.30) = 2 > 1,$$

so all inequalities invoked in the ledger and near-collision bounds hold with this concrete choice. Any neighbourhood of these values also works; we fix $(\sigma_S, \delta_D, \zeta; A, \gamma)$ as above for the compiled version.

A Appendix: A robust CLS route, kernel constants, and the pinned gate

Standing notation

Fix integers $X, H \in \mathbb{N}$ with $H \geq 1$ and write $N := H + 1$. Let $e : \mathbb{N} \rightarrow \mathbb{R}$ be the error sequence delivered by the analytic core. Define the windowed L^1 and L^2 quantities

$$A := \sum_{k=0}^H |e(X+k)|, \quad L^2 := \sum_{k=0}^H e(X+k)^2. \quad (\text{A.1})$$

Let S be a finite set of odd primes ($p \geq 3$) and define the truncated twin singular series

$$SS(S) := \prod_{p \in S} \frac{p(p-2)}{(p-1)^2} = \prod_{p \in S} \left(1 - \frac{1}{(p-1)^2}\right) \in (0, 1). \quad (\text{A.2})$$

Fix $\varepsilon \in (0, \frac{1}{2})$ and set $E := \varepsilon SS(S) \geq 0$.

Remark A.1 (No hidden cancellations). We never divide by N or by ε . Every square-root identity is used with explicit nonnegativity hypotheses; in particular $\sqrt{E^2} = E$ because $E \geq 0$, and $\sqrt{N}\sqrt{N/9} = N/3$ because $N \geq 1$.

Analytic input (L^2 with slack) and immediate consequences

[Windowed L^2 with slack] There exists a nonnegative function $E_{L^2}(X, H) \geq 0$ such that for all $X \geq X_0$,

$$L^2 \leq \varepsilon^2 SS(S)^2 \cdot \frac{N}{9} + E_{L^2}(X, H). \quad (\text{A.3})$$

When needed, assume a quantitative control

$$\frac{E_{L^2}(X, H)}{N} \leq \gamma \quad \text{for all } X \geq X_0, \quad (\text{A.4})$$

for some $\gamma \geq 0$.

Lemma A.2 (Windowed Cauchy–Schwarz). *With A, L^2, N as in (A.1), we have*

$$A \leq \sqrt{N} \sqrt{L^2}. \quad (\text{A.5})$$

Lemma A.3 (Elementary radical bounds). *If $a, b \geq 0$ and $a \leq b$, then $\sqrt{a} \leq \sqrt{b}$. Moreover, $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$.*

Lemma A.4 (Safe evaluation of the model radical). *Let $E = \varepsilon SS(S) \geq 0$ and $N \geq 1$. Then*

$$\sqrt{N} \cdot \sqrt{E^2 \cdot \frac{N}{9}} = E \cdot \frac{N}{3}. \quad (\text{A.6})$$

Theorem A.5 (CLS from L^2 with slack (CS route)). *Under Assumption A, uniformly for $X \geq X_0$,*

$$A \leq \varepsilon SS(S) \cdot \frac{N}{3} + \sqrt{N} \sqrt{E_{L^2}(X, H)}. \quad (\text{A.7})$$

Proof. By Lemma A.2 and (A.3),

$$A \leq \sqrt{N} \sqrt{E^2 \frac{N}{9} + E_{L^2}} \leq \sqrt{N} \left(\sqrt{E^2 \frac{N}{9}} + \sqrt{E_{L^2}} \right) = E \frac{N}{3} + \sqrt{N} \sqrt{E_{L^2}},$$

using Lemmas A.3 and A.4. □

Theorem A.6 (CLS from L^2 with slack (Young route)). *For any $t > 0$,*

$$A = \sum |e_k| \leq \frac{1}{2t} \sum e_k^2 + \frac{t}{2} \sum 1 = \frac{L^2}{2t} + \frac{Nt}{2}.$$

With (A.3) and $t^ = \sqrt{(E^2 N/9 + E_{L^2})/N}$,*

$$A \leq \sqrt{N} \sqrt{E^2 \frac{N}{9} + E_{L^2}} \leq E \frac{N}{3} + \sqrt{N} \sqrt{E_{L^2}},$$

matching (A.7) via an independent principle.

Corollary A.7 (Multiplicative slack). *If $E_{L^2}(X, H) \leq \gamma E^2 \cdot \frac{N}{9}$ for some $0 \leq \gamma \leq 1$, then*

$$A \leq (1 + \sqrt{\gamma}) \varepsilon SS(S) \cdot \frac{N}{3}. \quad (\text{A.8})$$

Corollary A.8 (Averaged slack). *If $E_{L^2}(X, H) = o(N)$ uniformly in X , then*

$$\frac{A}{N} \leq \varepsilon SS(S) \cdot \frac{1}{3} + o(1). \quad (\text{A.9})$$

Kernel, TT* to slack, and explicit constants

We model the TT* tail by a kernel $K = w * \tilde{w}$, where $\tilde{w}(n) = w(-n)$. Assume for every shift t , a dispersion bound $\Xi(t) \leq M$.

Lemma A.9 (TT* to slack). *If $L^2 \leq E^2 \frac{N}{9} + \sum_{t \in \mathbb{Z}} |K(t)| \Xi(t)$ with $\Xi(t) \leq M$, then*

$$E_{L^2} \leq M \|K\|_{\ell^1(\mathbb{Z})}.$$

*If moreover $K = w * \tilde{w}$ and $w \in \ell^1$, then $\|K\|_{\ell^1} \leq \|w\|_{\ell^1}^2$. If $w \geq 0$, both bounds are equalities in the worst case.*

Proof. Immediate from $\sum_t |K(t)| \Xi(t) \leq M \sum_t |K(t)|$ and the ℓ^1 Young inequality $\|w * \tilde{w}\|_1 \leq \|w\|_1 \|\tilde{w}\|_1 = \|w\|_1^2$. □

Explicit nonnegative window (tent/Fejér). Define

$$w(n) = \begin{cases} 1 - \frac{n}{H}, & 0 \leq n \leq H, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{nonnegative on } [0, H])$$

Then

$$\|w\|_{\ell^1} = \sum_{n=0}^H \left(1 - \frac{n}{H}\right) = \frac{H+1}{2}, \quad \|K\|_{\ell^1} \leq \|w\|_1^2 = \frac{(H+1)^2}{4}.$$

Hence, by Lemma A.9,

$$E_{L^2} \leq M \frac{(H+1)^2}{4}, \quad \gamma = \frac{E_{L^2}}{N} \leq M \frac{H+1}{4}. \quad (\text{A.10})$$

This improves the flat window ($\gamma \lesssim MH$) by a factor ≈ 4 , with a fully rigorous, sign-robust bound.

Finite singular-series floor

For any finite $S \subset \{p \geq 3\}$,

$$SS(S) = \prod_{p \in S} \left(1 - \frac{1}{(p-1)^2}\right) \geq \left(\frac{3}{4}\right)^{\#S} =: s_0 > 0. \quad (\text{A.11})$$

If one wishes a concrete example, take $S = \{3, 5, 7\}$:

$$s_0 = \frac{3}{4} \cdot \frac{15}{16} \cdot \frac{35}{36} = \frac{1575}{2304} \approx 0.6840,$$

and hence $SS(S) \geq s_0$ is fully explicit.

Pinned majors, leakage, and the gate

Let $W_k \geq 0$ denote the model main weight per shift and write

$$\text{TwinWeight}_k = W_k + e_k. \quad (\text{A.12})$$

[Pinned majors] There exists $c_0 > 0$ such that

$$\sum_{k=0}^H W_k \geq c_0 SS(S) N, \quad (\text{A.13})$$

and a leakage parameter $\eta \geq 0$ such that

$$\text{Leakage} \leq \eta c_0 SS(S) N. \quad (\text{A.14})$$

[Finite Goldbach TI] All local constraints required by the major/minor synthesis hold for $X \geq X_0$.

Proposition A.10 (Pinned gate). *If Assumptions A–A hold and*

$$\sum_{k=0}^H |e_k| \leq \theta c_0 SS(S) N \quad (0 \leq \theta < 1 - \eta), \quad (\text{A.15})$$

then

$$\sum_{k=0}^H \text{TwinWeight}_k \geq (1 - \theta - \eta) c_0 SS(S) N > 0, \quad (\text{A.16})$$

hence some k has $\text{TwinWeight}_k > 0$.

Theorem A.11 (Finite-conditional twin in the window). *Assume A, A, A . Fix $s_0 > 0$ with $SS(S) \geq s_0$ and take γ as in (A.4) (or (A.10)). If*

$$\frac{\varepsilon}{3c_0} + \frac{\sqrt{\gamma}}{c_0 s_0} + \eta < 1, \quad (\text{A.17})$$

then for every $X \geq X_0$ there exists $k \in [0, H]$ with $\text{TwinWeight}_k > 0$; in the standard normalization this yields a twin in $[X, X + H]$.

Proof. By Theorem A.5 (or A.6) and (A.4),

$$\sum_k |e_k| = A \leq \frac{\varepsilon}{3} SS(S) N + \sqrt{N} \sqrt{E_{L^2}} \leq \left(\frac{\varepsilon}{3} + \frac{\sqrt{\gamma}}{s_0} \right) SS(S) N.$$

Thus (A.15) holds with $\theta = \frac{\varepsilon}{3c_0} + \frac{\sqrt{\gamma}}{c_0 s_0}$, and (A.17) ensures $\theta < 1 - \eta$. Apply Proposition A.10. $\square$

Corollary A.12 (Infinitely many twins under uniform hypotheses). *If the hypotheses of Theorem A.11 hold uniformly for all sufficiently large X (or along an infinite sequence $X \rightarrow \infty$) with fixed $(c_0, s_0, \varepsilon, \gamma, \eta)$ satisfying (A.17), then there are infinitely many twin primes.*

Deterministic leakage bound (prime powers)

In $[X, X + H]$, the number of squares is $\leq \lceil \sqrt{X + H} - \sqrt{X} \rceil \leq H/(2\sqrt{X}) + 1$; higher powers contribute even less. With log-weights,

$$\text{Leakage} \ll \left(\frac{H}{\sqrt{X}} + 1 \right) \log X.$$

Hence for $H \leq X^{1/2-\delta}$ one can fix $\eta = \eta(\delta, X_0)$ that tends to 0 as $X_0 \rightarrow \infty$, and use it in (A.17).

Robustness lemmas (window changes, shifts)

Lemma A.13 (Shift stability). *If Assumptions A–A hold uniformly in X , then replacing X by $X + \Delta$ with $|\Delta| \leq H$ preserves all bounds with the same constants.*

Lemma A.14 (Length perturbation). *Let H' satisfy $|H' - H| \leq \Delta$. Then*

$$A' \leq A + \Delta \cdot \frac{E}{3} + \sqrt{\Delta} \sqrt{E_{L^2}^*},$$

where $E_{L^2}^ = \sup_{X, h \in [H, H']} E_{L^2}(X, h)$. In particular, $E_{L^2} = o(N)$ remains $o(N)$ under bounded relative variations of H .*

Uniformity audit (explicit quantifiers)

There exist X_0 , finite $S \subset \{p \geq 3\}$, and constants $\varepsilon \in (0, \frac{1}{2})$, $\gamma \geq 0$, $c_0 > 0$, $\eta \geq 0$ such that for every $X \geq X_0$ and every $H \geq 1$ in the chosen regime:

$$L^2(X, H) \leq \varepsilon^2 SS(S)^2 \frac{N}{9} + E_{L^2}(X, H), \quad 0 \leq \frac{E_{L^2}(X, H)}{N} \leq \gamma,$$

$$\sum_{k=0}^H W_k \geq c_0 SS(S) N, \quad \text{Leakage} \leq \eta c_0 SS(S) N,$$

GTI holds (finite interface).

All constants may depend on S and bank parameters, but not on X or H .

Safe operating regime (one-line gate check)

Specializing (A.17) with the tent window bound (A.10) and any explicit singular-series floor s_0 (e.g. (A.11) or $S = \{3, 5, 7\}$),

$$\boxed{\frac{\varepsilon}{3c_0} + \frac{\sqrt{M(H+1)/4}}{c_0 s_0} + \eta < 1} \quad (\text{A.18})$$

is a sufficient numerical condition for a twin in $[X, X + H]$, uniformly for all $X \geq X_0$.

Adversarial stress tests (tightness)

- **CS extremizer.** Concentrating e on one index gives $A = \sqrt{N}\sqrt{L^2}$. Both the CS and Young routes recover this extremal scaling, so (A.7) is tight against concentration.
- **Gate adversary.** If $\sum |e_k| = \theta c_0 S S N$ with $\theta \uparrow 1 - \eta$, then $\sum (W_k + e_k) \downarrow 0$; the margin inequality is sharp by construction.

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